Technische Hochschule Rosenheim

Speech Processing

Speech Processing

What to expect:

- basic concepts of speech and audio processing
- speech signal processing
- basics of human speech production
- speech recognition
- automatic speech recognition
- speech classification

Speech is Spoken Language! It is not just audible text!

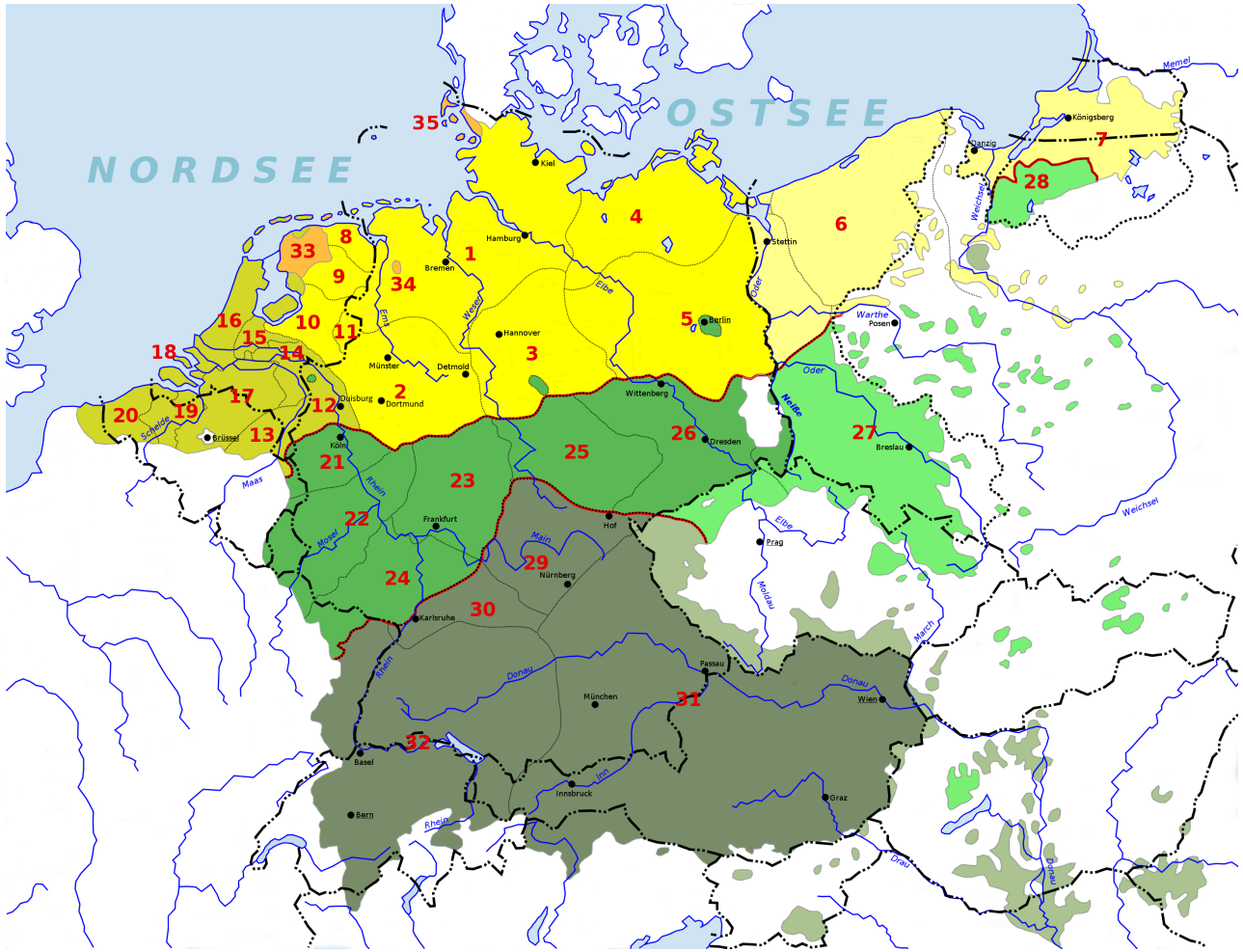
Speech is ...

- ambiguous
- effortful
- contrastive
- prosodic
- adaptive
- context-dependent
- meaningful
- referential
- indexical
- rhetorical
- personalized
- affective
- multimodal
- contaminated

Today's Agenda

- Sound
- Speaking
- Hearing
- Phonetics
- Phonology
- Prosody
- Behavior

Properties of Spoken Language

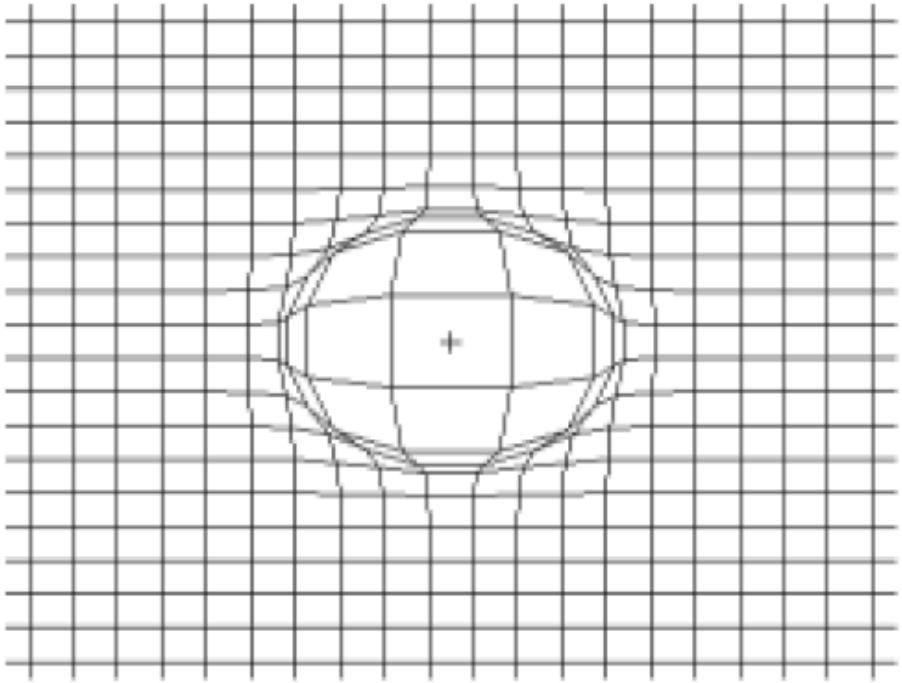


https://www.dialekte.schule.bayern.de/sprachatlas/

https://www.bbc.co.uk/voices/

What are Sounds?

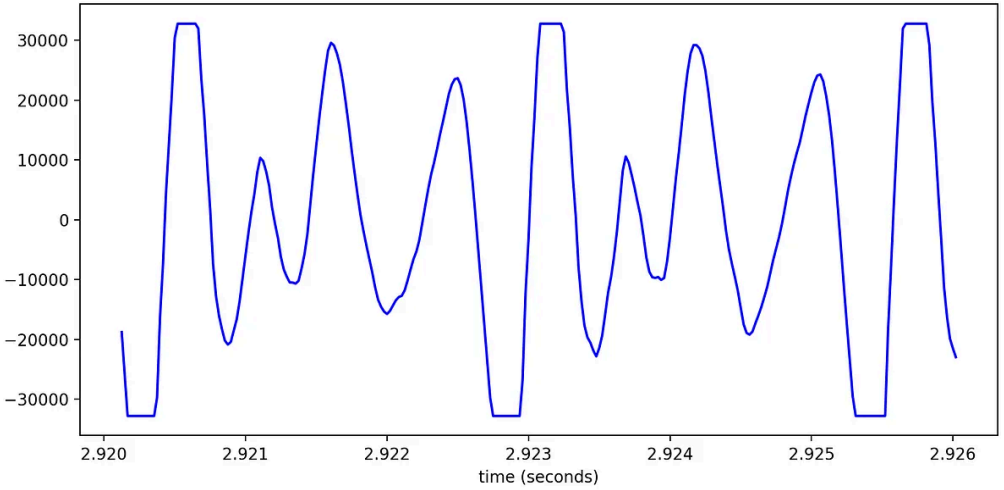
- acoustic energy
- mechanical waves
- vibrations, moving through the air
- variations in air pressure / density
- **Sound travels in waves**
  - longitudinal waves
  - compressions and rarefactions
  - **speed of sound: 343 m/s (in air)**



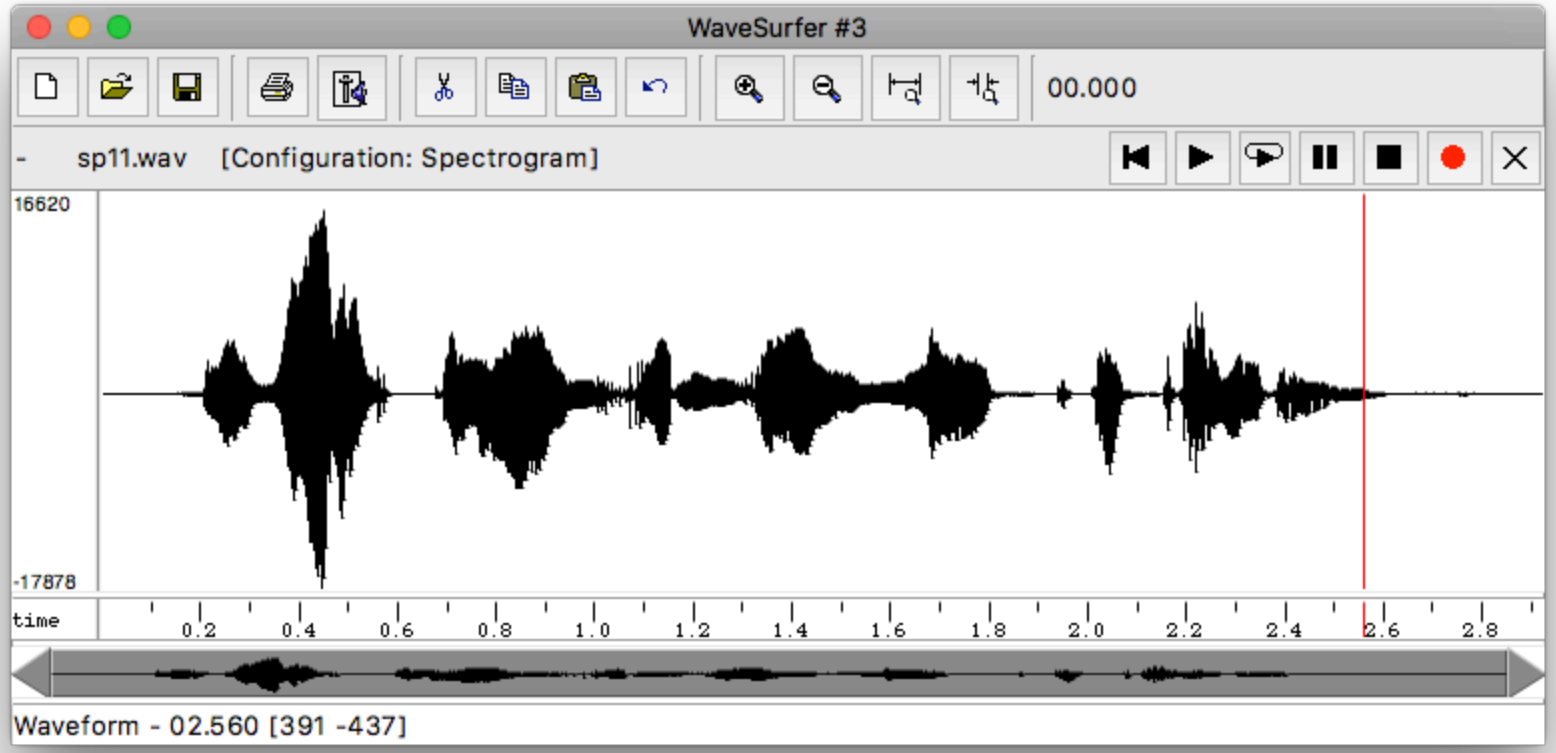
Recording Sound

- a **microphone** is an electrical device
  - converts acoustic signals to electrical energy.
- The diaphragm is a thin flexible material
  - vibrates back and forth when picking up oscillating air particles of a sound wave
- Microphones have a minimum and maximum amplitude they can handle
  - largely determines their performance/ quality
- Pre-amplifiers are used to boost the signal
  - before it is digitized by an analog-to-digital converter (ADC) which samples the signal at a fixed rate
- if the signal is too loud, it will be clipped
  - the top and bottom of the waveform are cut off
  - this is called **clipping**

Example fo clipping in a 6ms cut from a waveform:



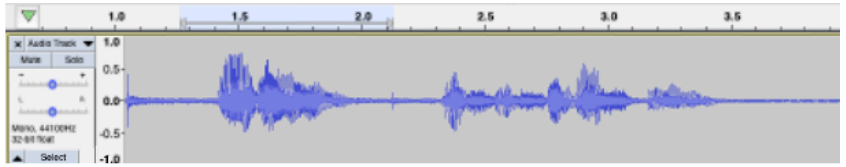
Voice Recording

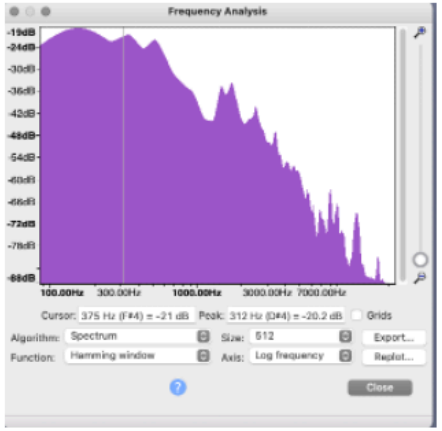


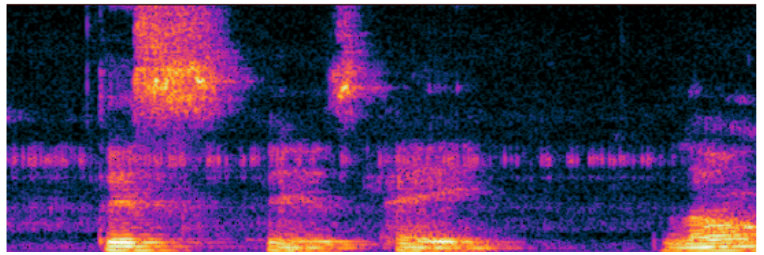
0:00 / 0:02

"Measuring" Sound

- **Amplitude** (loudness)
  - measured in decibels (dB)
- **Frequency** (pitch)
  - measured in Hertz (Hz)
- **Waveform**
  - Time vs. Amplitude
- **Spectrum**:
  - Amplitude vs. Frequency
- **Spectrograms**
  - visual representation of the spectrum of frequencies of a signal as it varies with time
  - x-axis: time
  - y-axis: frequency
  - color: amplitude



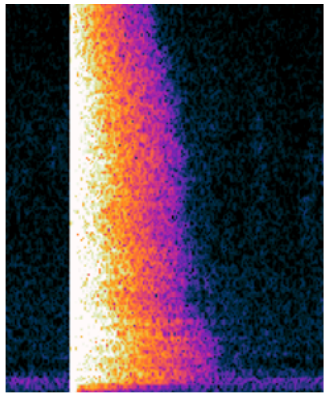


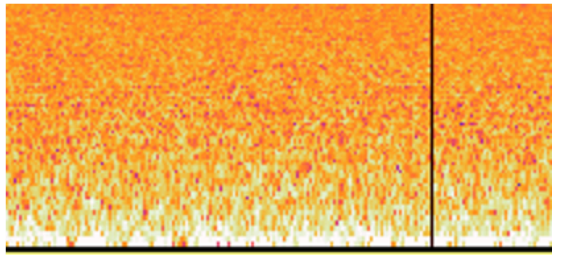


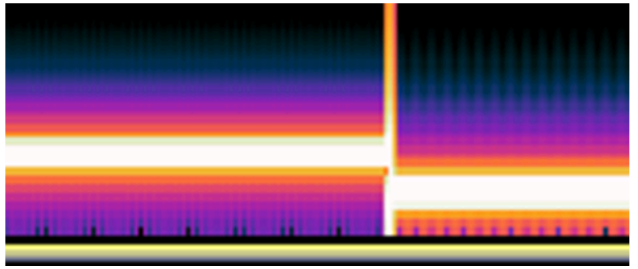
Properties of Sound

Reflect how sound pressure can vary over time

- **Impulsive**
  - energy across the frequency spectrum
  - for a short time
  - e.g., a clap, a click, a pop
- **Noisy**
  - random, unpredictable
  - multiple frequencies and energy levels
  - across the frequency spectrum
  - e.g., a waterfall, a crowd, the wind
- **Repetitive**
  - energy on a few frequencies
  - and their harmonics
  - e.g., a whistle, a siren, a bell, instrumental music







Discretization of Sound: Effects of the Sampling Rate

200Hz

0:00 / 0:09

500Hz

0:00 / 0:09

1000Hz

0:00 / 0:09

2000Hz

0:00 / 0:09

4000Hz

0:00 / 0:09

5000Hz

0:00 / 0:09

6000Hz

0:00 / 0:09

7000Hz

0:00 / 0:09

Nyquist Theorem and Sampling Rate

Sampling Rate / Frequency

- rate at which an analog signal is sampled to convert it into a digital signal
- Measured in samples per second (Hz).

Nyquist Theorem

- Fundamental principle in the field of digital signal processing.
- a continuous signal can be properly sampled and reconstructed if it is sampled at a rate greater than twice its highest frequency component.
- **Nyquist Rate**: minimum sampling rate required to avoid aliasing, equal to twice the highest frequency of the input signal.
- **Aliasing**: Distortion that occurs when a signal is undersampled, causing different signals to become indistinguishable

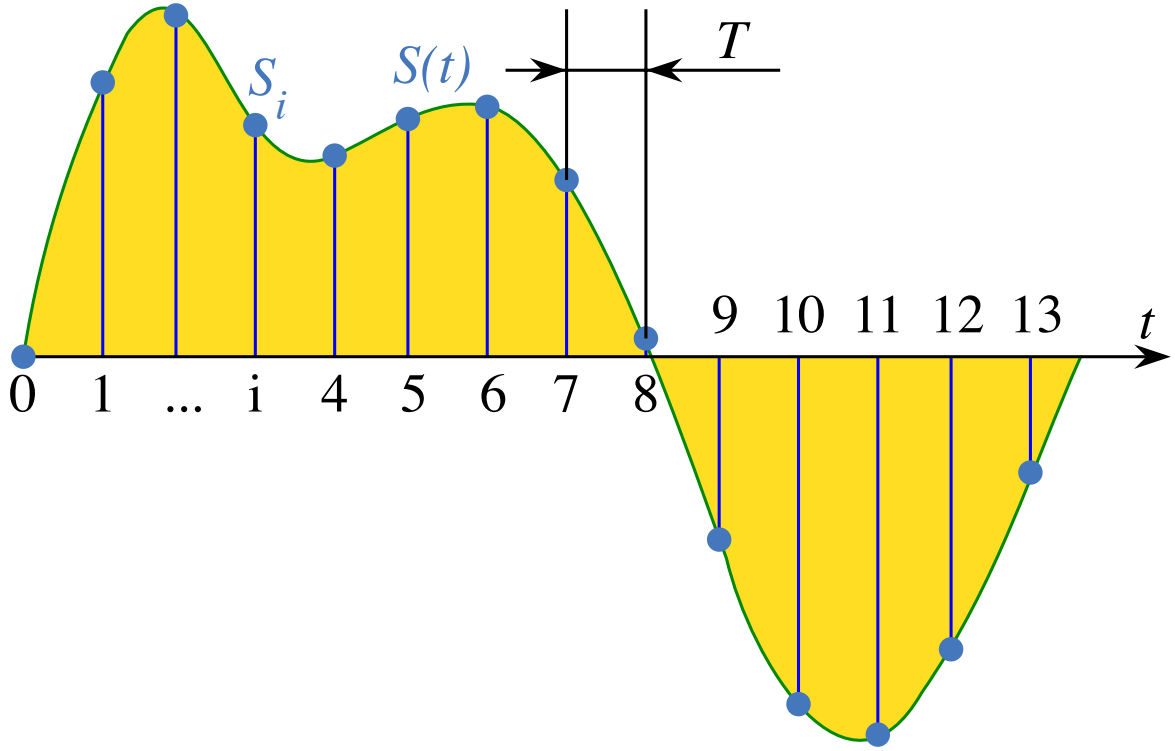


Image source: By Д.Ильин: vectorization - File:Signal Sampling.png by Email4mobile (talk), CC0, https://commons.wikimedia.org/w/index.php?curid=98587159

Nyquist Theorem and Sampling Rate

- **Nyquist Rate Formula**: $(f_s > 2B)$
  - Where f_s is the sampling rate and B is the highest frequency of the signal.
- **Alias Frequency Formula**: $f_{alias} = |f_s - f|$
  - Where f is the original signal frequency.

Examples Music CD / Telephone Speech :

CD

- standard sampling rate 44.1 kHz, sufficient for the human hearing range (up to 20 kHz).
- Higher sampling rates are used in professional audio to capture more detail.

Telephone

- Sampled at 8kHz
- Human fundamental frequency range: $\approx$ 85-255 Hz
- Harmonics (multiples of those frequencies) can extend over 5 kHz
- $\rightarrow$ no beautiful singing voice over the phone

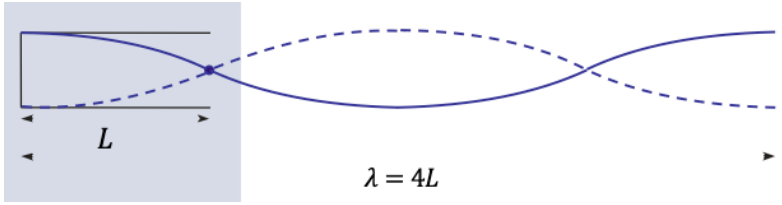
Source-Filter Model

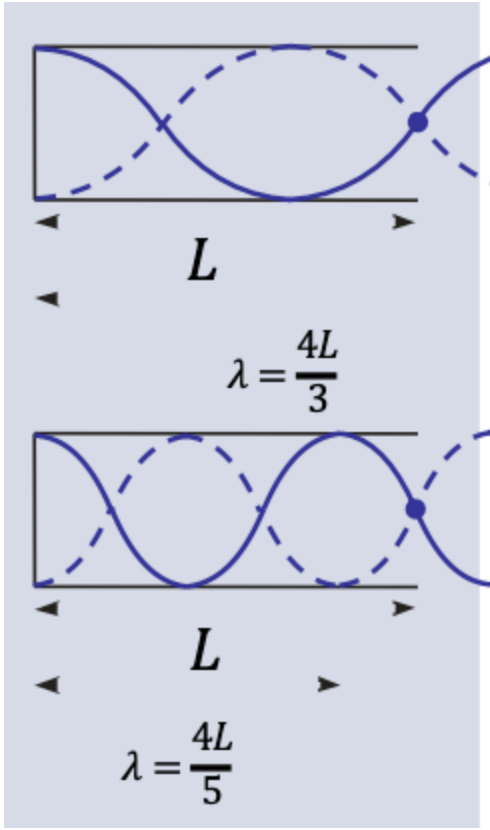
- A vibration source generates a "source signal."
- Partial oscillations at various frequencies
- Generate **resonances** in an appended tube
    - Depending on the shape, certain frequencies are amplified or attenuated
- destructive interference results in **anti-resonances**
    - reduced amplitude of certain frequencies
- The filters can be shaped with a tube system
- The tubes thus function as a **filter**

Source–filter Model of Speech Production

- speech is viewed as the result of passing a glottal excitation signal (source)
- through a time-varying linear filter (resonant characteristics of the vocal tract)
- known source–filter system is that based on linear prediction (LP) of speech.
- In its simplest form, a time-varying filter modeled as an autoregressive (AR) filter is excited by either quasiperiodic pulses (during voiced speech), or noise (during unvoiced speech).
- Many attempts have been made to improve the source (excitation) signal in the LP context.
- This includes multipulse LP and code-excited linear prediction (CELP).
- A more-compact and at the same time flexible representation of the excitation signal has been proposed from a family of speech representations referred to as sinusoidal models (SM).
- In SM, the excitation signal for both voiced and unvoiced speech is modeled.

Resonant Structures





The resonant frequency f of a cylinder closed at one end, and open at the other is given by:

$$f = \frac{(2n-1)v}{4L}$$

The Human Vocal Tract

- In the neutral position ("uh"), the human vocal tract is equivalent to a tube closed at one (the larynx) end and open at the other (the mouth)
- The resonant frequencies are, therefore, given by the previous formula.
- For an average adult male, the length of the vocal tract is about 17.5cm. The resonances are:
  - fundamental frequency: 490 Hz
  - 1st harmonic 1470 Hz
  - 2st harmonic 2450 Hz
  - 3rd harmonic 3430 Hz

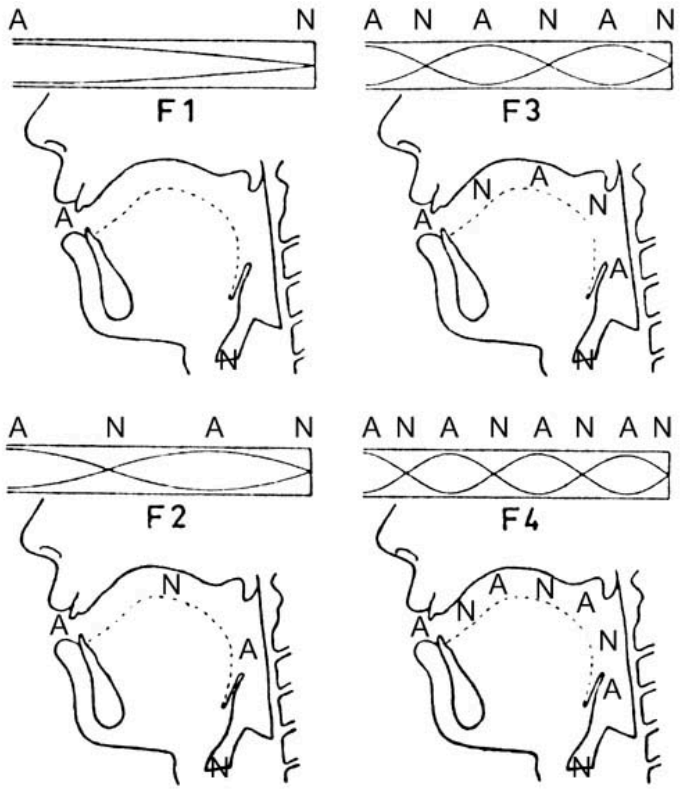


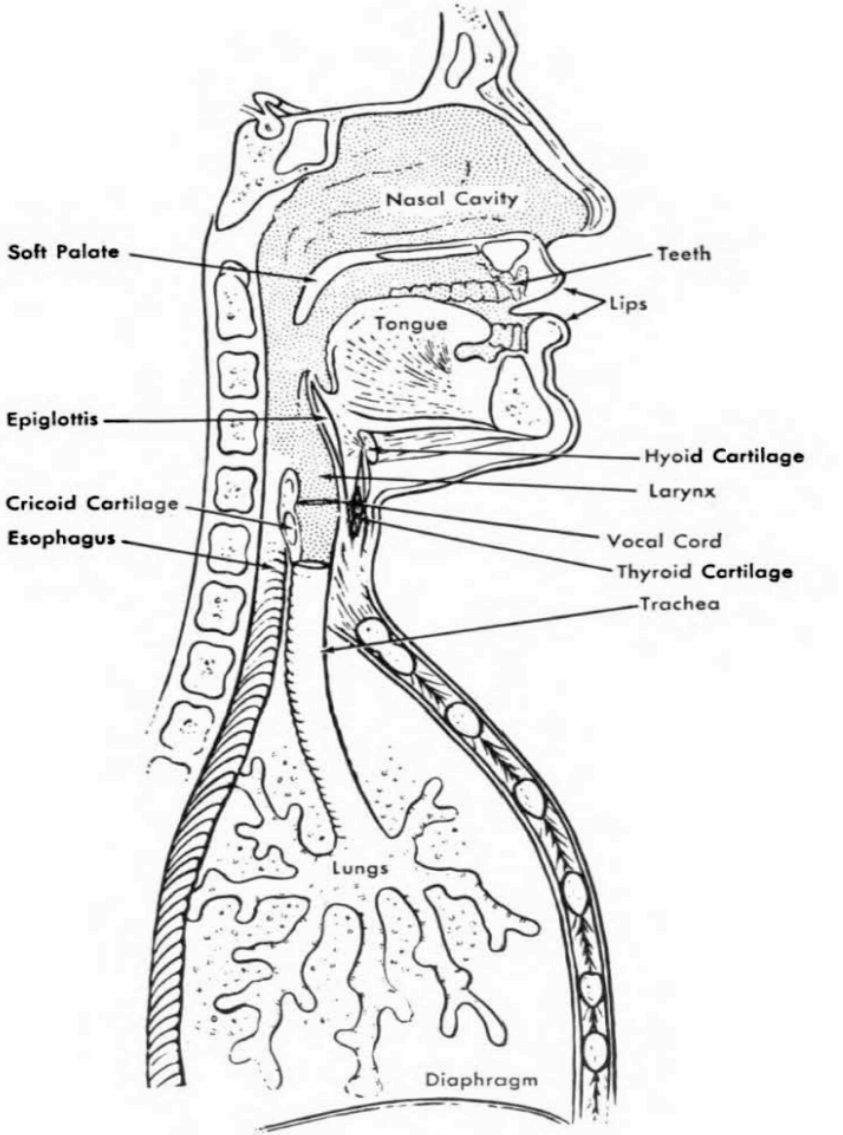
Image source: Chiba & Kajiyama (1958)

WARNING

It is easy to confuse the fundamental frequency of a repetitive sound source with the fundamental frequency of a resonant structure.

These are two different things!

The Human Vocal Organs



Vocal apparatus evolved for...

- Breathing
- Eating
- Vocalizing
- Speaking

Main source of energy:

- Lungs

Main Source of sound:

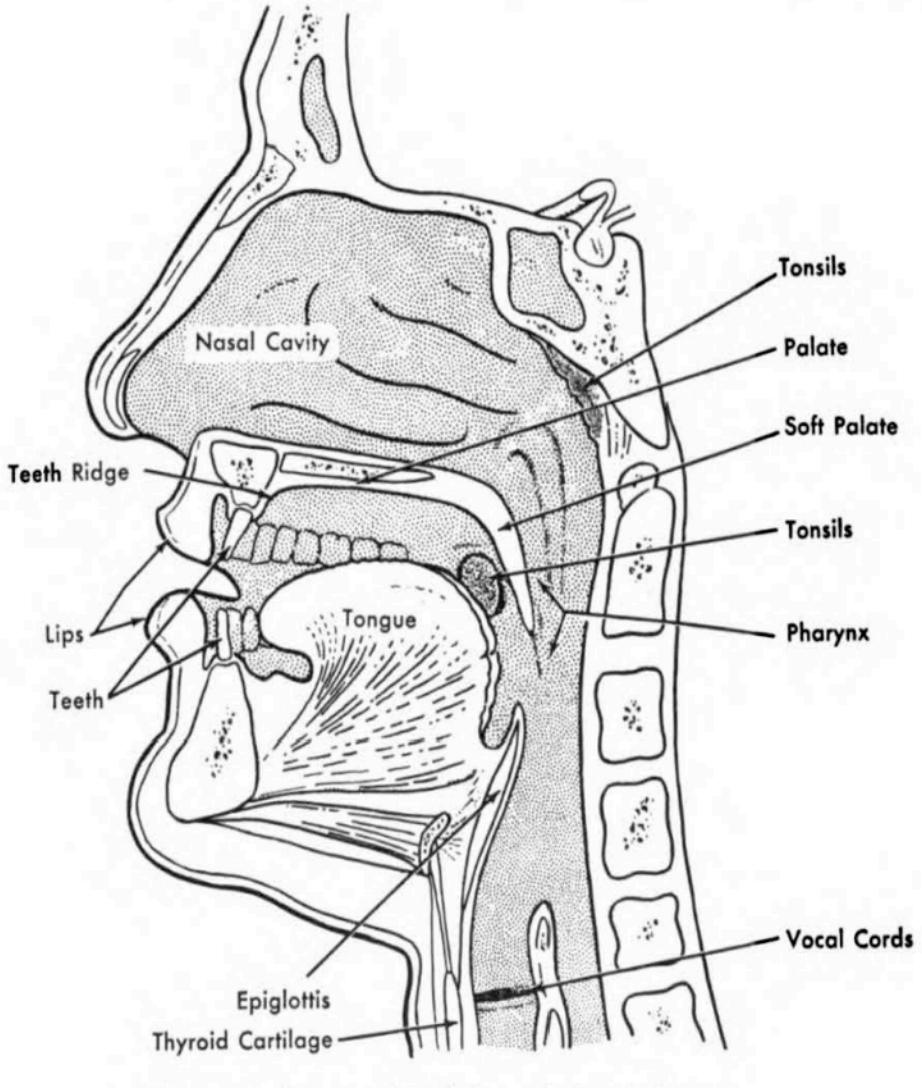
- Vocal cords

Main source of articulation:

- Tongue

Image source: Denes, P.B., Pinson, E.N., 1993. The speech chain: the physics and biology of spoken language.

The Human Vocal Tract



Voice Source:

- Vocal cords

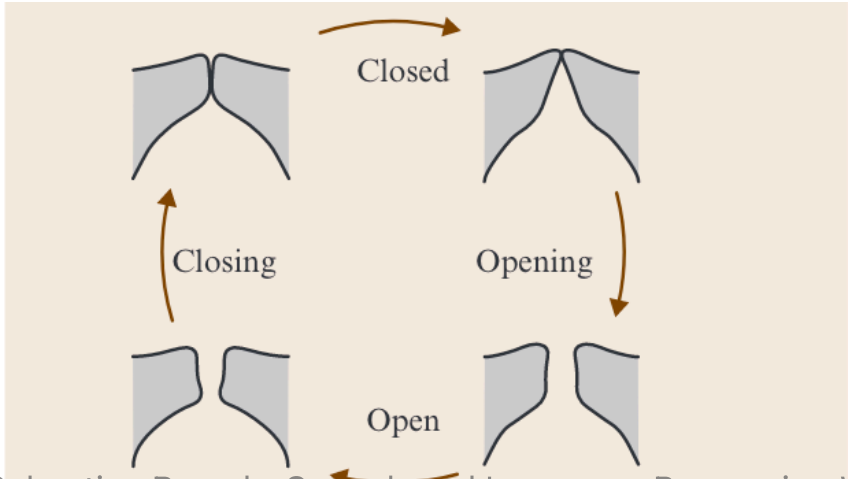
Filter:

- Vocal tract

Image source: Denes, P.B., Pinson, E.N., 1993. The speech chain: the physics and biology of spoken language.

Voice Source

- Air pressure from the lungs builds up behind closed **vocal cords** or **vocal folds**
- The vocal folds are repeatedly forced and pulled together again, producing a series of small pulses of air
- This modulation of the airstream is known as **phonation**
- The tension in the muscles attached to the vocal folds determines their rate of vibration and hence the **fundamental frequency** (and perceived pitch) of the voice
- Because the vibration is not a pure tone, there is energy at the fundamental frequency and its **harmonics**



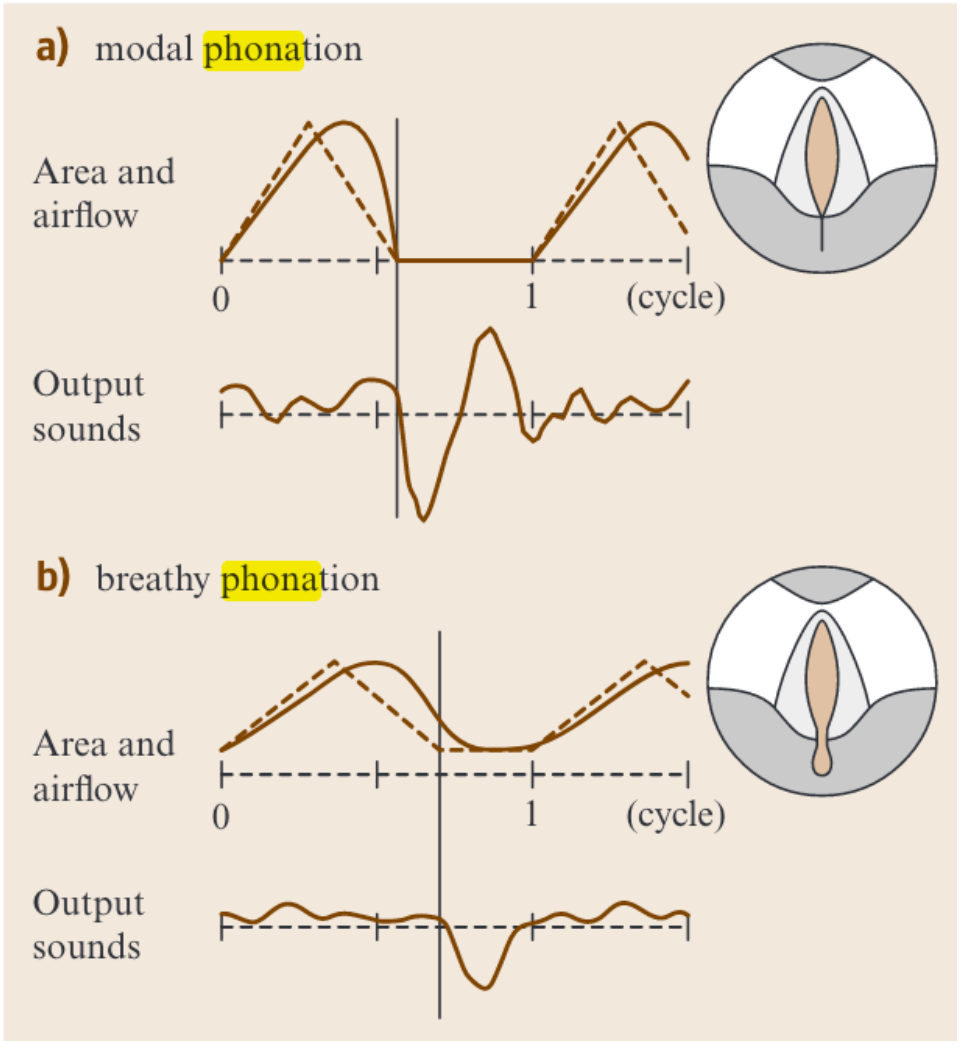


Image source: Benesty, J., Sondhi, M.M., Huang, Y.A. (Eds.), 2008. Springer Handbook of Speech Processing, Springer Handbooks. Springer Berlin Heidelberg, Berlin, Heidelberg.

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The Human Larynx

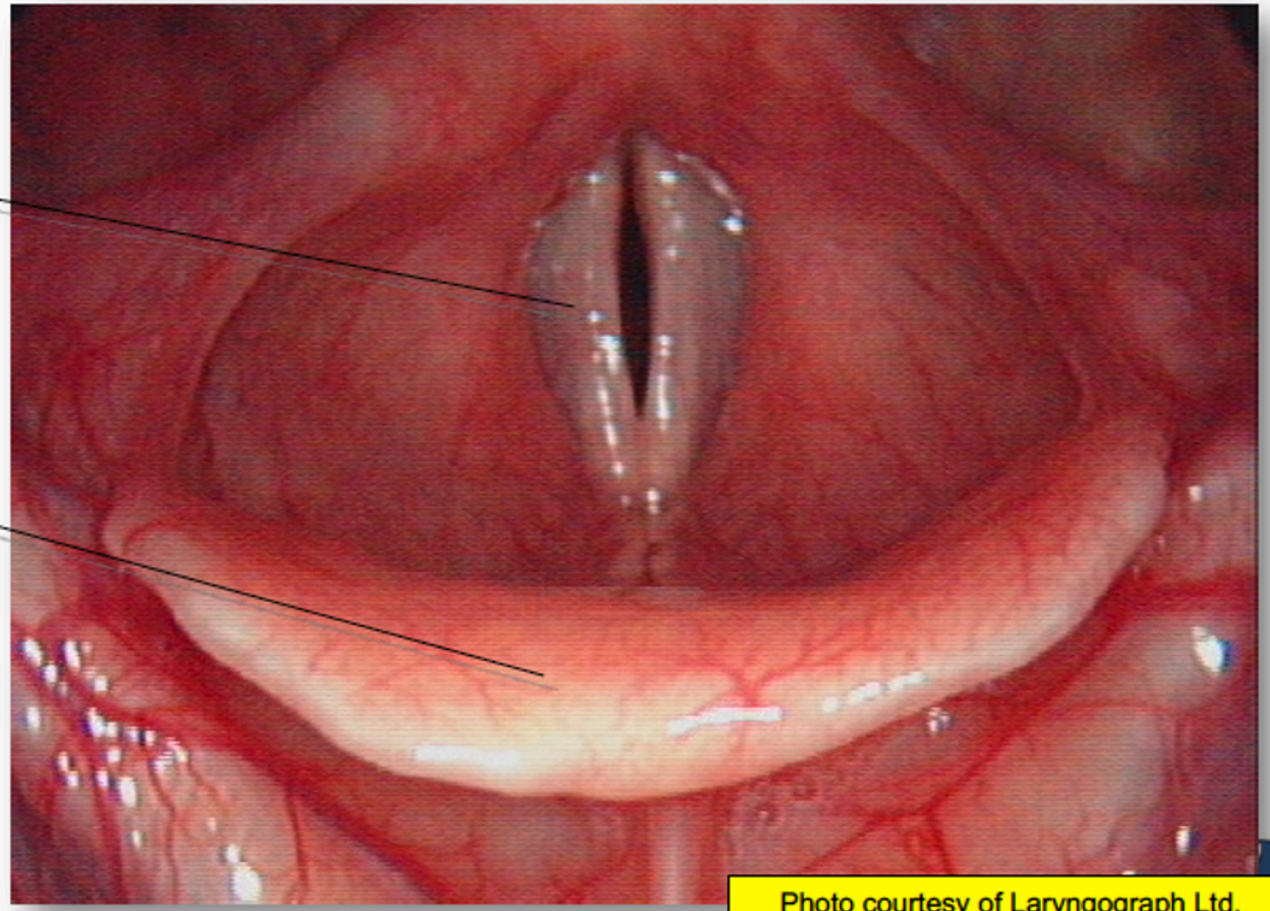
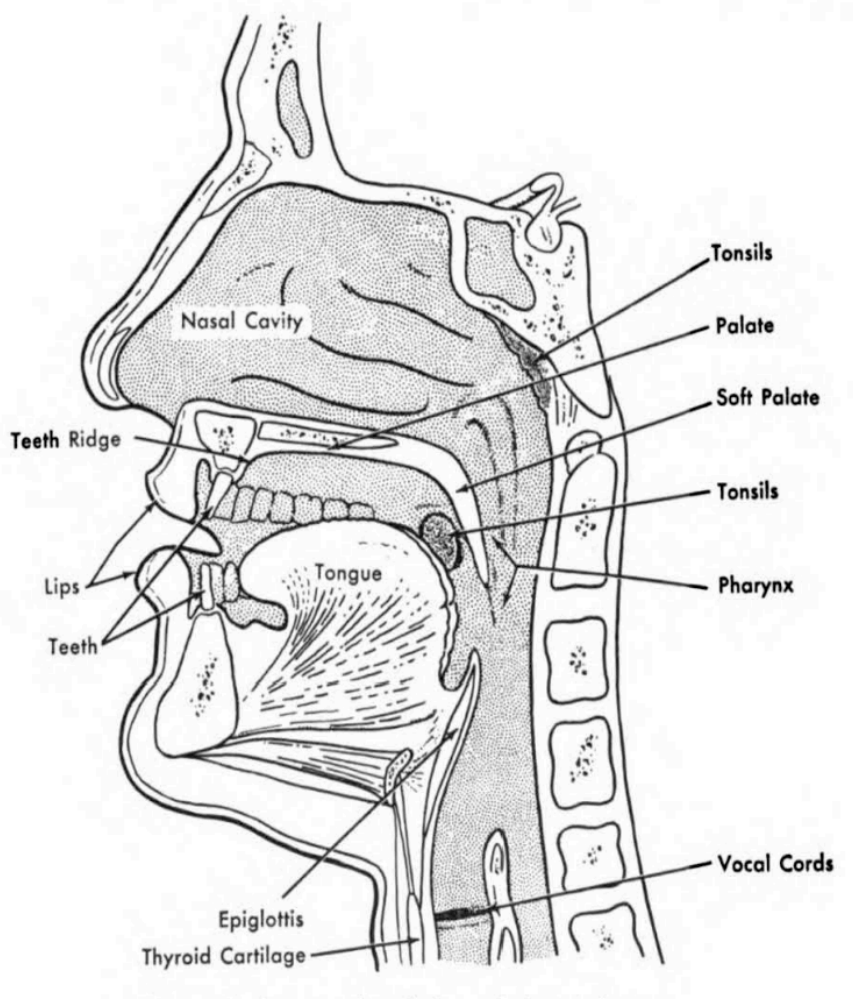
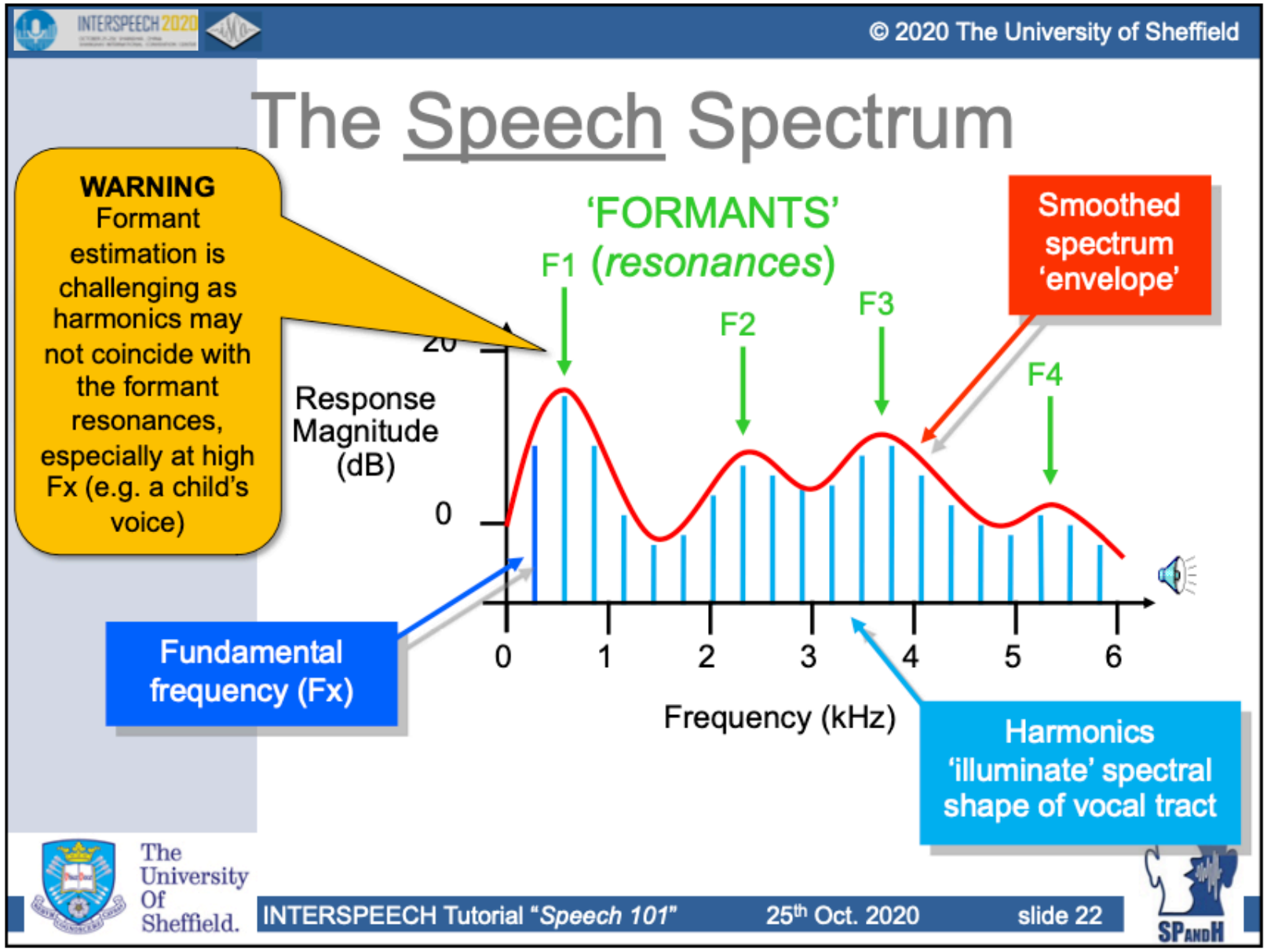


Photo courtesy of Laryngograph Ltd.

Voice Filter

- The vocal tract forms a **resonator** with a complex shape
- Resonances are known as **formants**
- Speech is produced by using the **articulators** to change the shape of the vocal tract, hence modifying its resonant characteristics
- Different configurations of the vocal tract enhance some of the harmonics of the pitch, and suppress (damp) others
- The principal articulator is the **tongue**, but the jaw, lips, soft palate and teeth are also involved

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The Speech Spectrum
WARNING
Formant estimation is challenging as harmonics may not coincide with the formant resonances, especially at high Fx (e.g. a child's voice)
'FORMANTS'
F1 (resonances)
F2
F3
F4
Smoothed spectrum 'envelope'
Response Magnitude (dB)
20
0
0
1
2
3
4
5
6
Frequency (kHz)
Fundamental frequency (Fx)
Harmonics 'illuminate' spectral shape of vocal tract
The University Of Sheffield.
INTERSPEECH Tutorial "Speech 101"
25th Oct. 2020
slide 22
SPANDH

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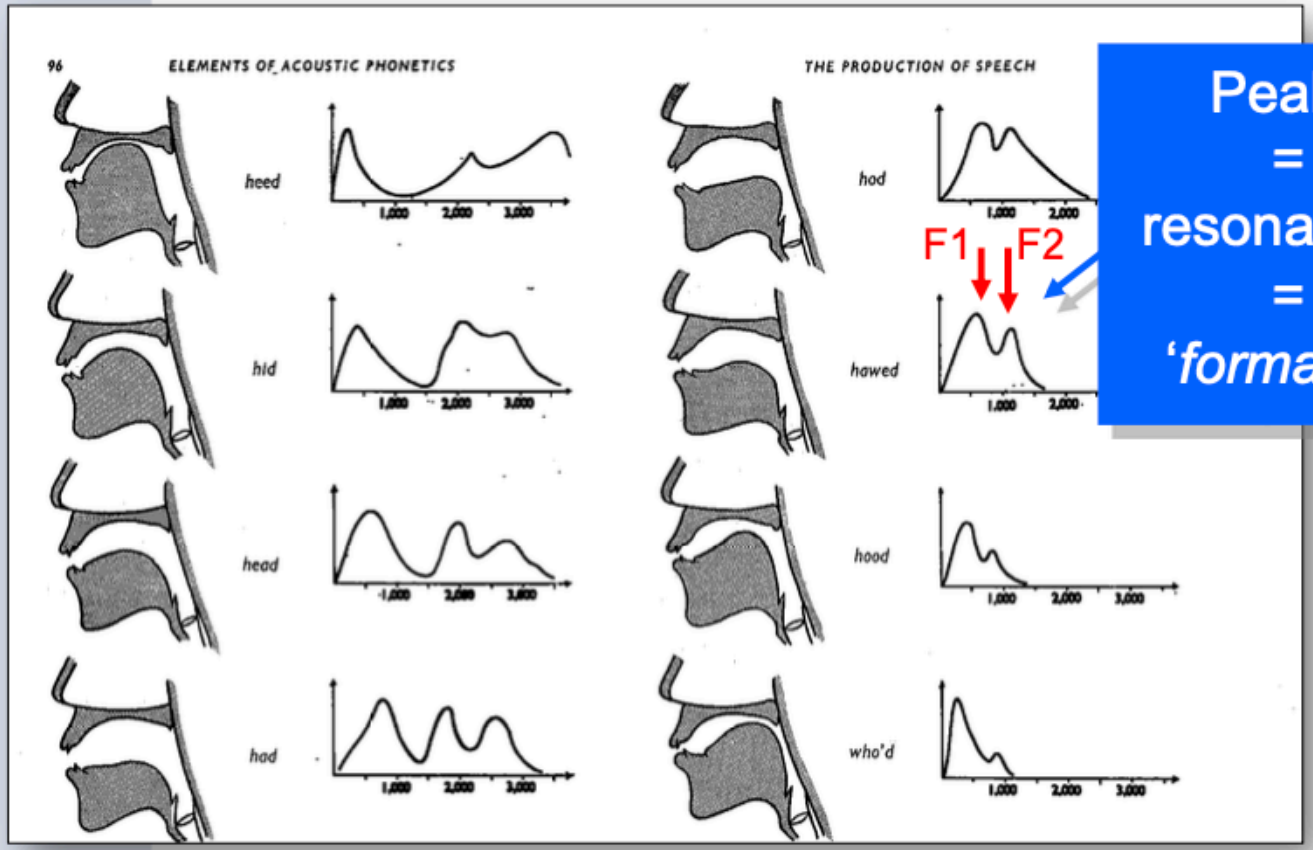
Vocal Tract Shape and Spectra



Peaks = resonances = *'formants'*

Taken from: Ladefoged, P. (1962). ***Elements of Acoustic Phonetics***: London: University of Chicago Press.

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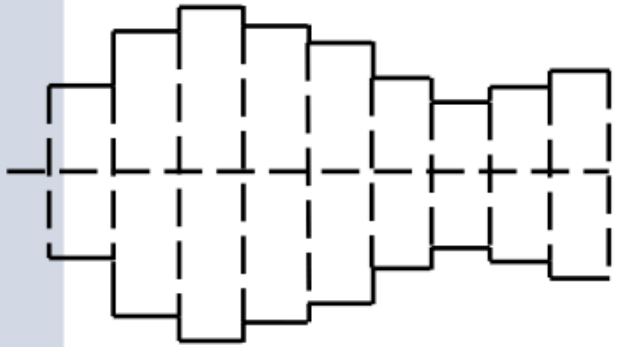
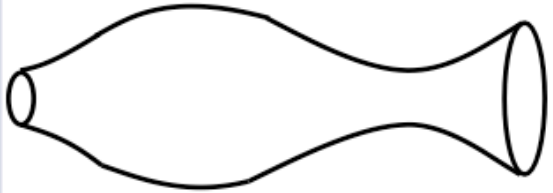
Simulating Vocal Tract Shapes



Glottis Lips

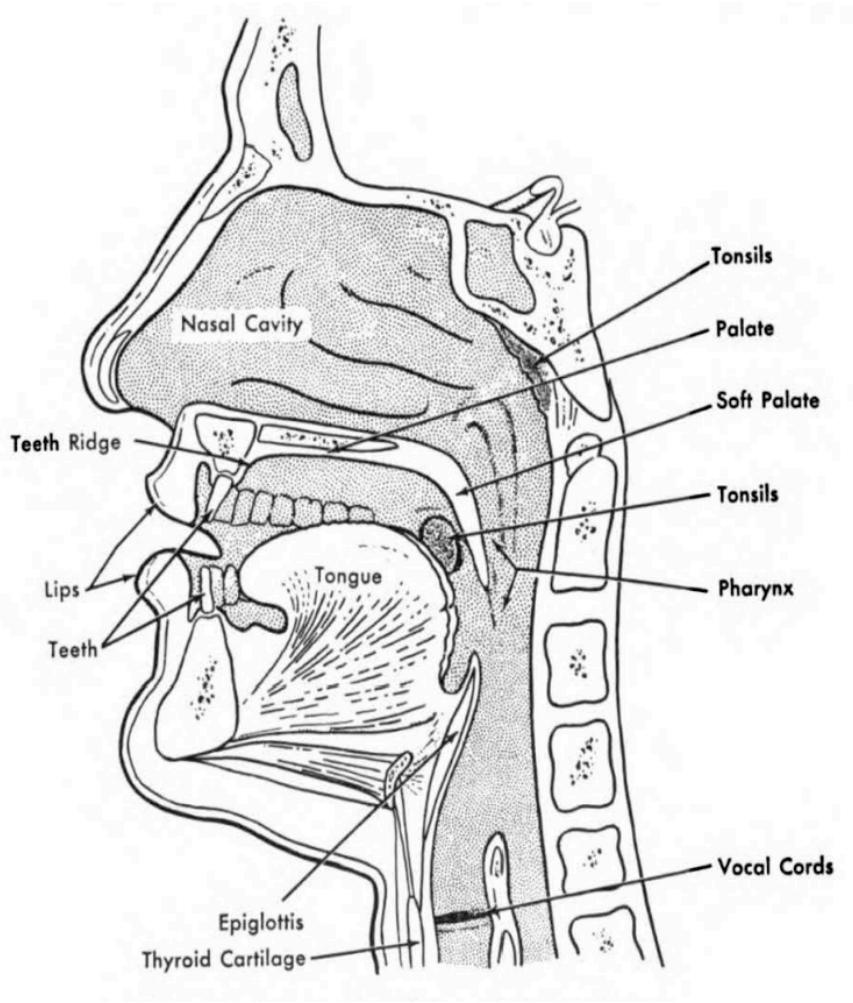


3D printed vocal tracts courtesy of Takayuki Arai

Adapted from: Vary, P. & Martin, R. (2006), *Digital Speech Transmission*, Wiley.

Sound Sources in the Vocal Tract

- The vibration of the vocal folds is not the only source of sound in the human vocal tract
- A **noisy** sound can be generated anywhere there is a partial constriction causing turbulent air flow (e.g. “sh”)
- An **impulsive sound** can be generated by releasing a blockage (e.g. at the lips)
- A **repetitive sound** can be generated by vibrating an articulator (e.g. the tongue)
- A **pure tone** (e.g. a whistle) can be generated by exciting a resonance



Beatboxing Video

Articulation

Voice

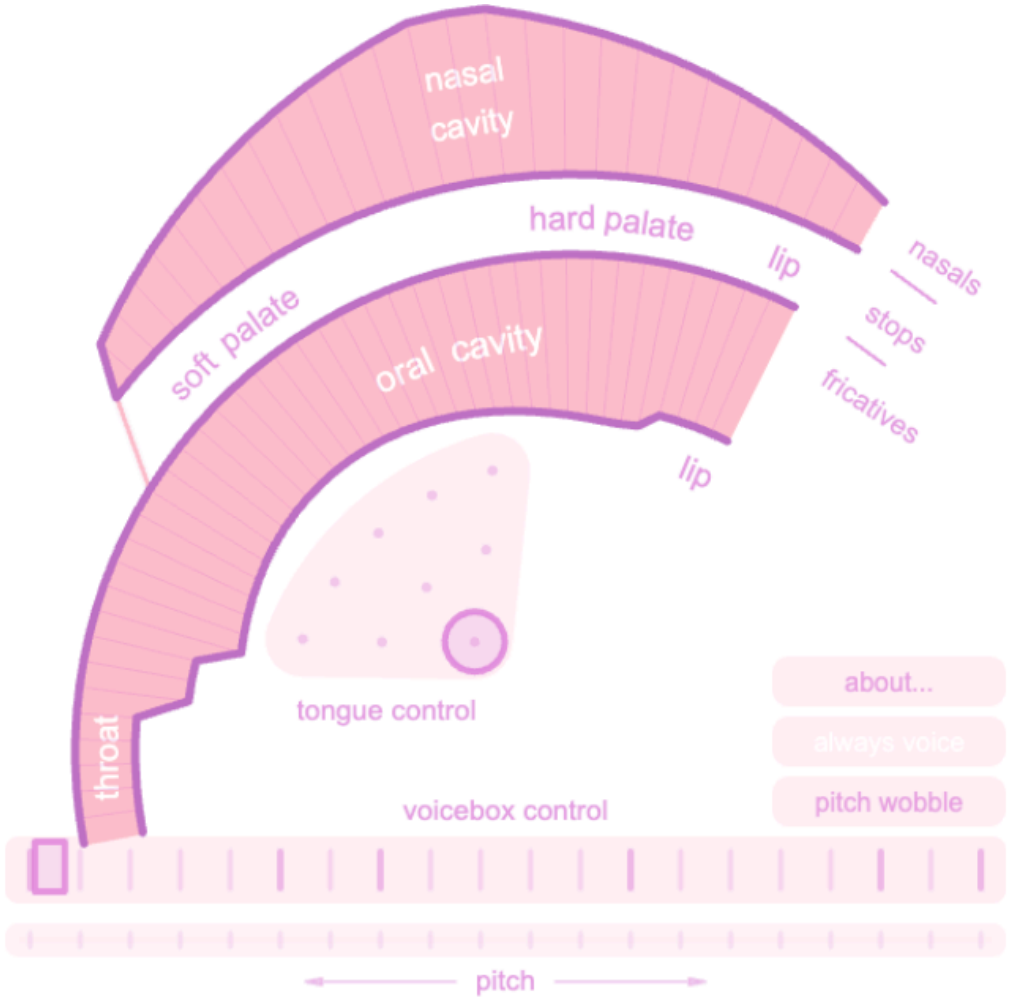
- level of voicedness (voiced-voiceless)
- Voice quality (creaky, breathy, modal)

Manner

- manner of the airflow modification
- by the primary and secondary articulators

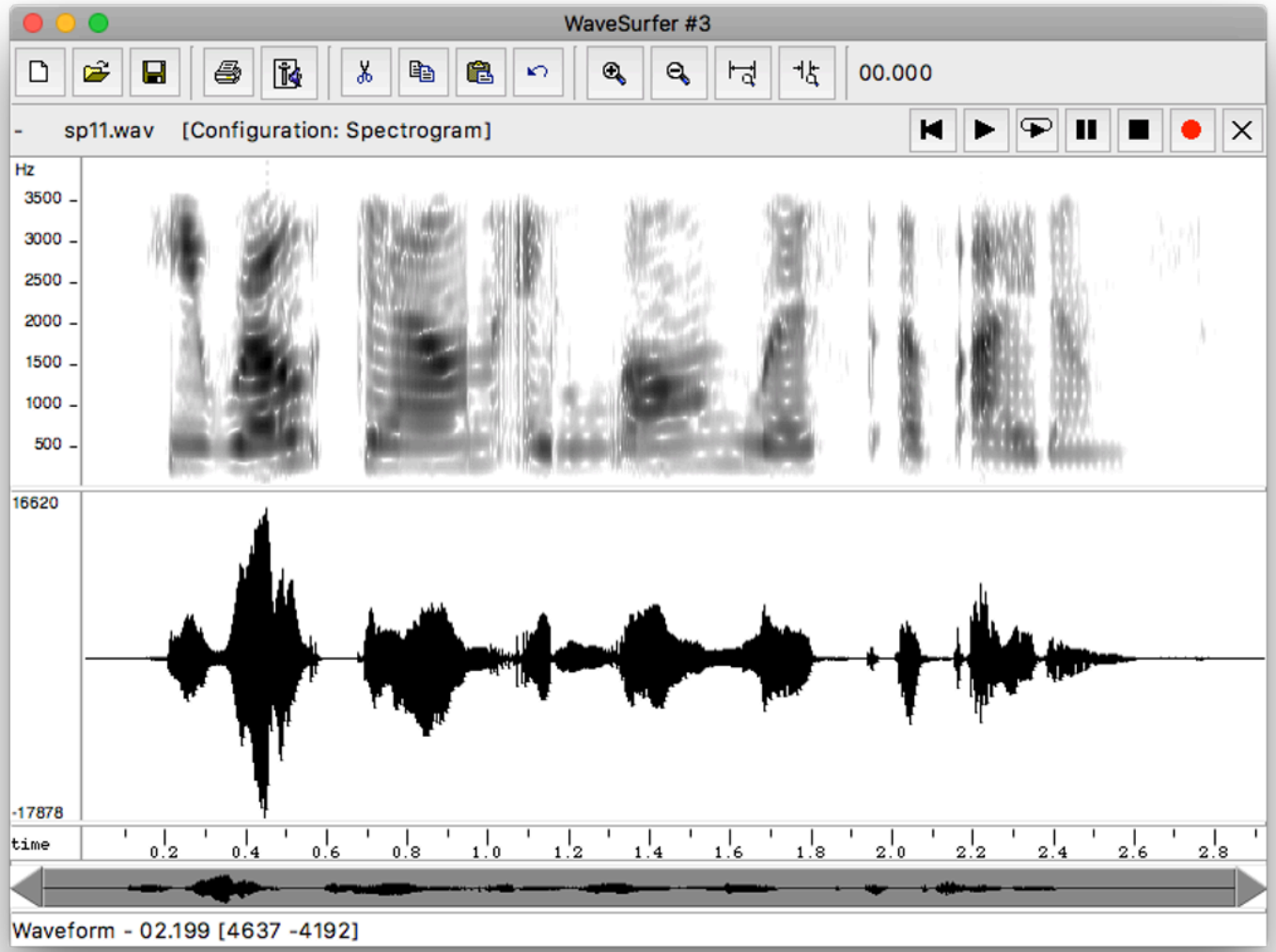
Place

- Physical location of sound production in the vocal tract

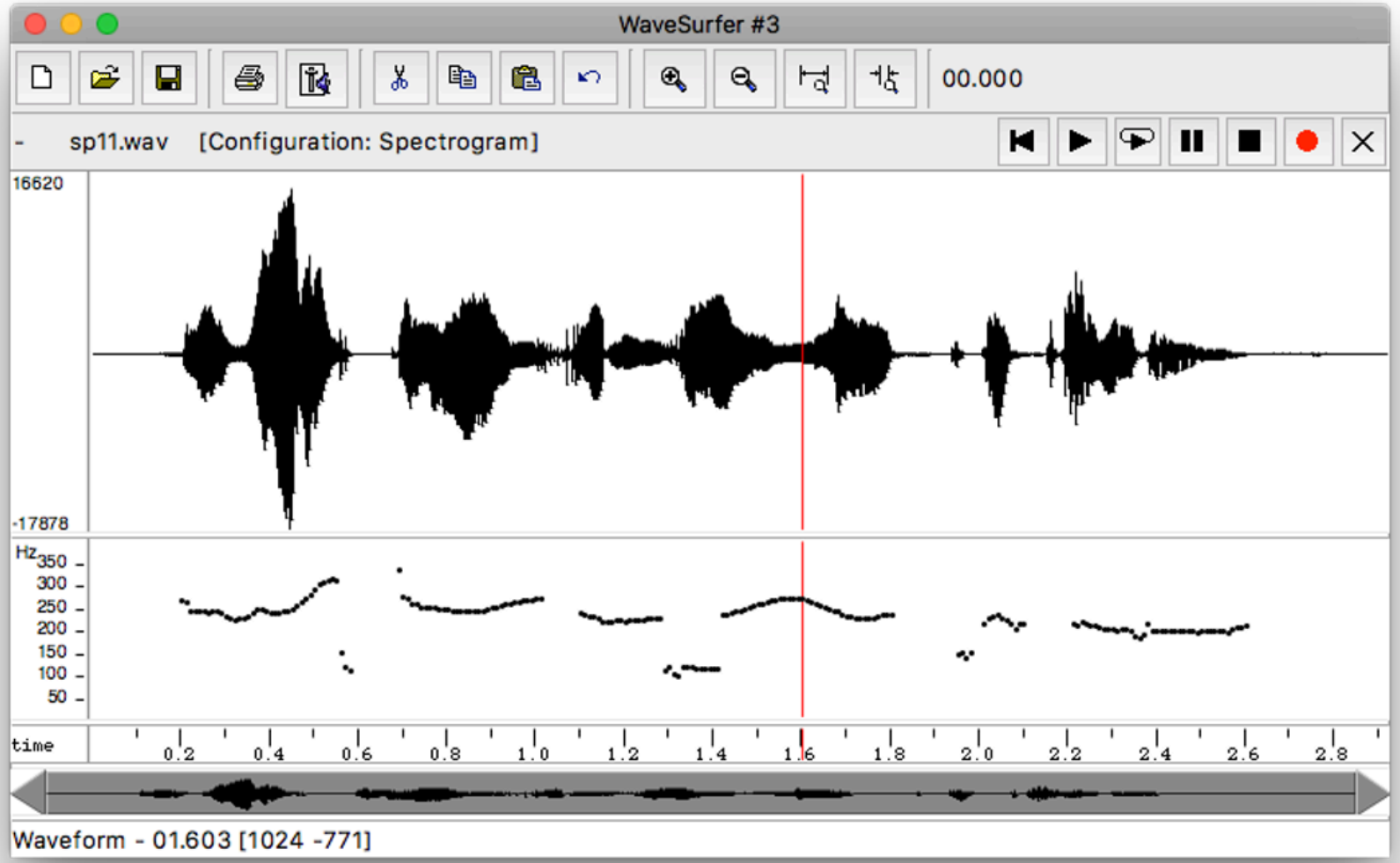


Demo: https://dood.al/pinktrombone/

Voice Frequencies



Fundamental Frequencies



Fundamental Frequency (F0) / Pitch in male and female speakers

Table 1. *Mean f_0 (Hz) measured on the 27 (C)VCV words for female (n=5) and male (n=5) French speakers and female (n=5) and male (n=5) American English speakers. Standard deviation among the 135 measurements (27 words * 5 speakers) is also mentioned for the four groups.*

	French Speakers		American speakers	
	Females	**Males**	**Females**	**Males**
Mean f_0 (Hz) - all words	**234**	**133**	**210**	**119**
SD	*18*	*12*	*27*	*19*

Image source: Erwan Pépiot. Male and female speech: a study of mean f0, f0 range, phonation type and speech rate in Parisian French and American English speakers. Speech Prosody 7, May 2014, Dublin, Ireland. pp.305-309. ffhalshs-00999332f

The Human Ear

- The auditory system has evolved for:
  – acoustic sensing
    - sound localization
    - – communication
- Primary function: **frequency analysis**
- The main percepts are:
    - **pitch** (for a repetitive sound)
    - **loudness** (log related to the sound pressure level)
    - **timbre**(related to the energy at different frequencies)

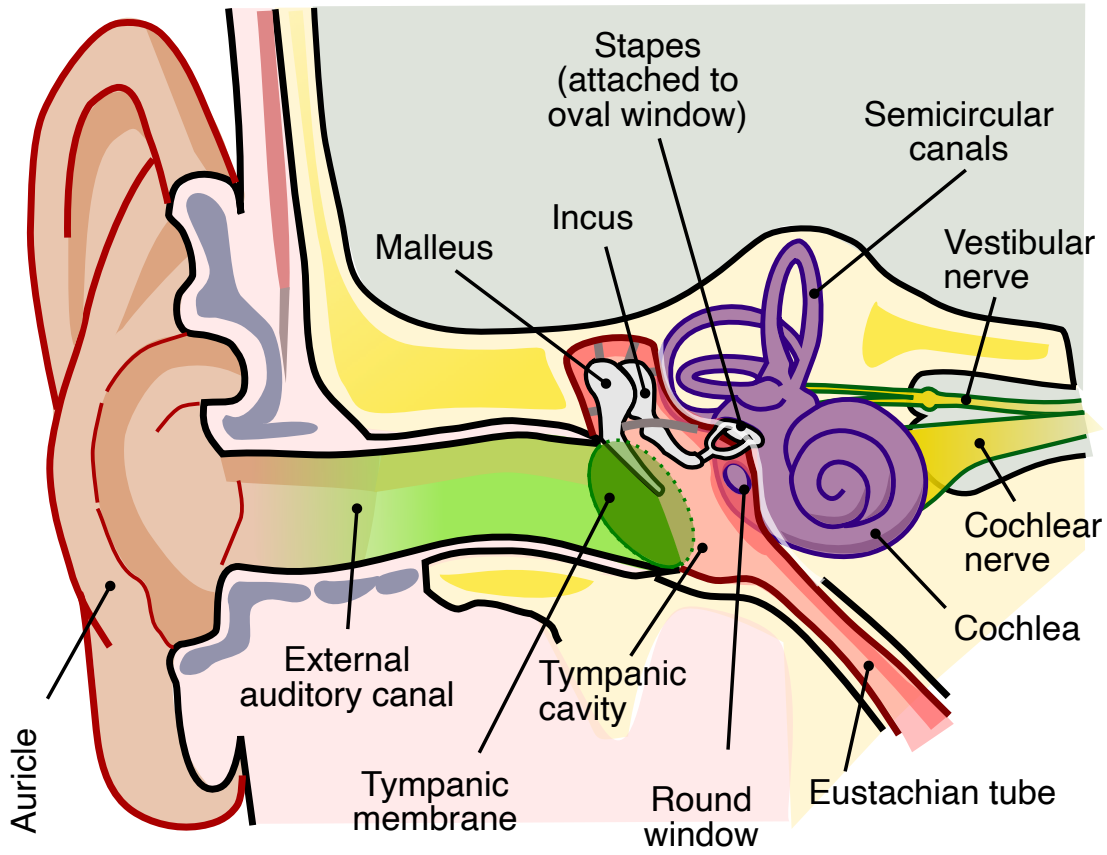


Image source: https://en.wikipedia.org/w/index.php?lang=en&title=File%3AAnatomy_of_the_Human_Ear.svg

The Human Ear

Outer Ear

- directional sensitivity
- visible part of the ear

Middle Ear

- air-to-liquid transduction
- mechanical amplification
- and overload protection

Inner Ear

- place-depended liquid-to-neural transduction
- equivalent to a bank of bandpass filters

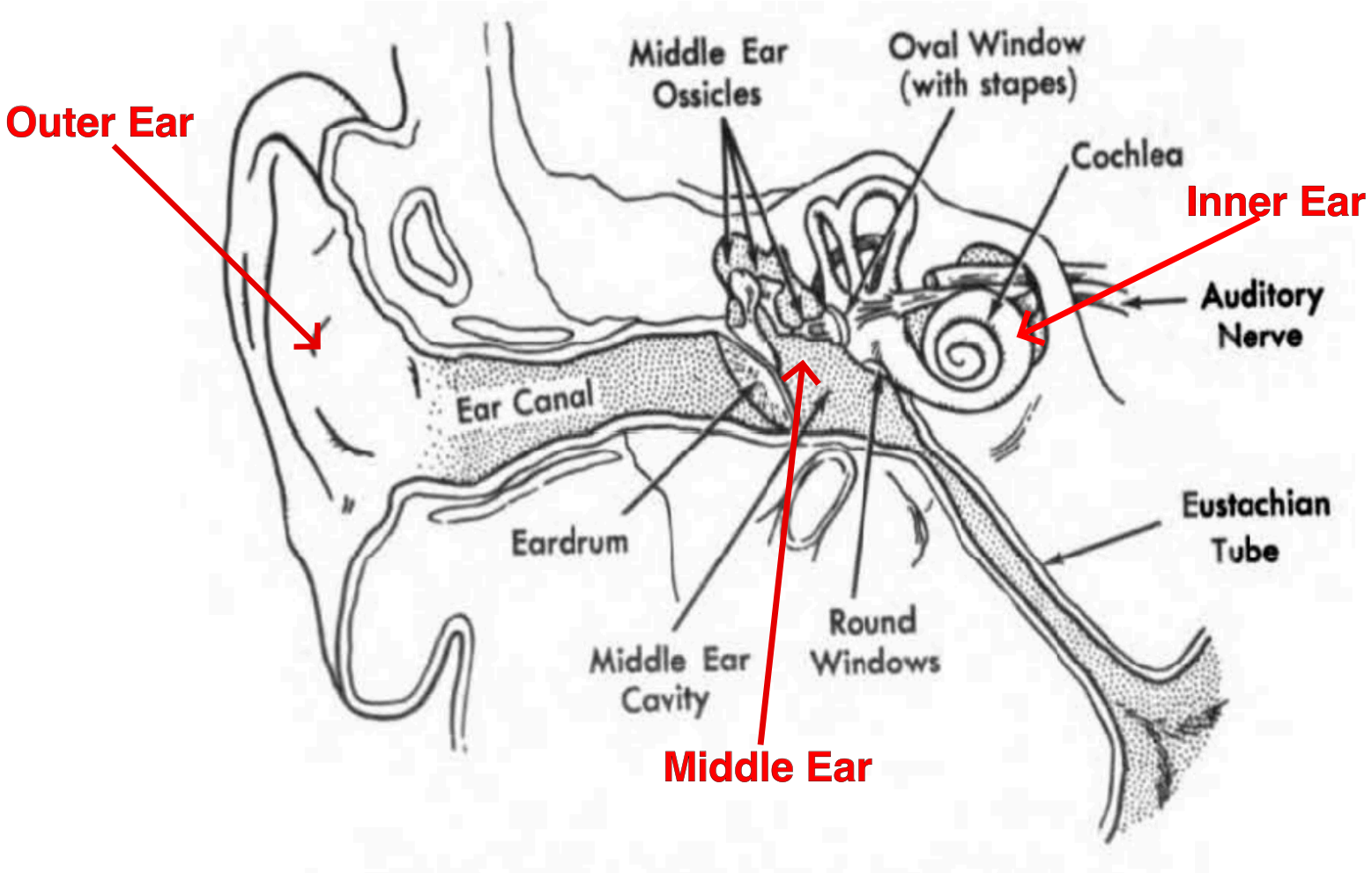
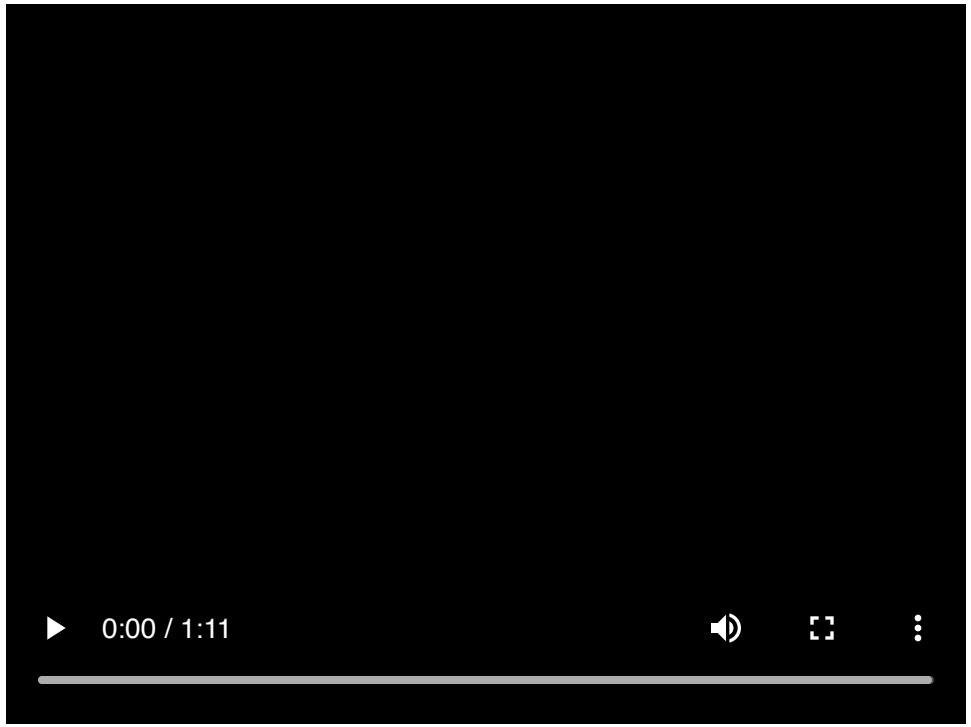
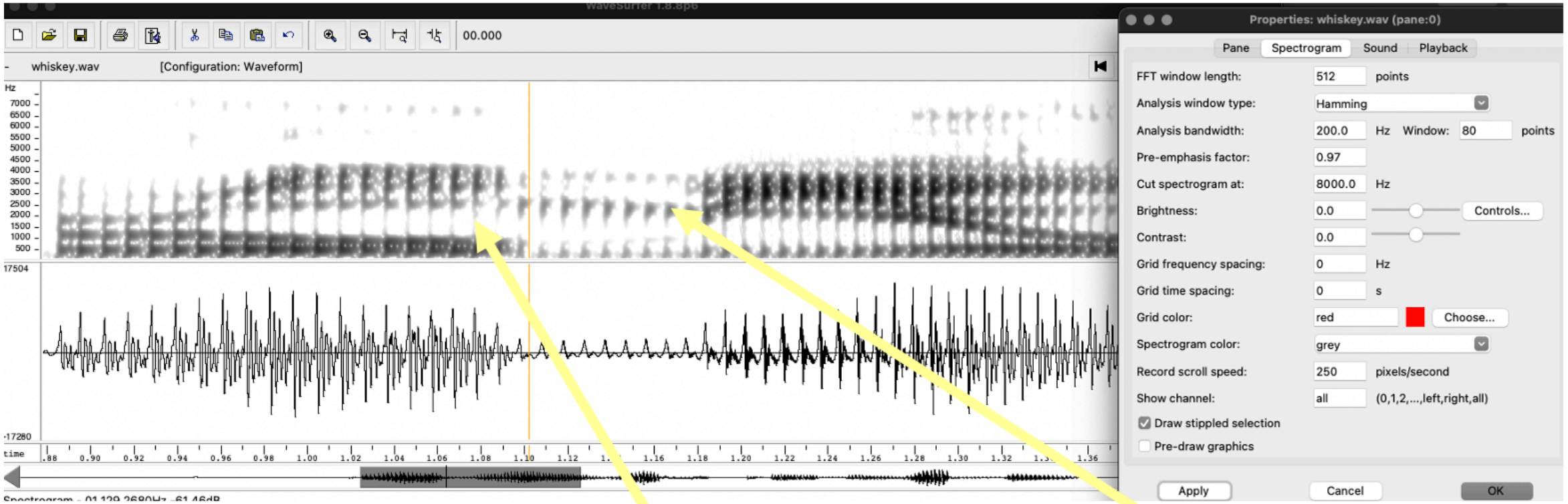


Image source: Denes, P.B., Pinson, E.N., 1993. The speech chain: the physics and biology of spoken language

Chochlea in Action



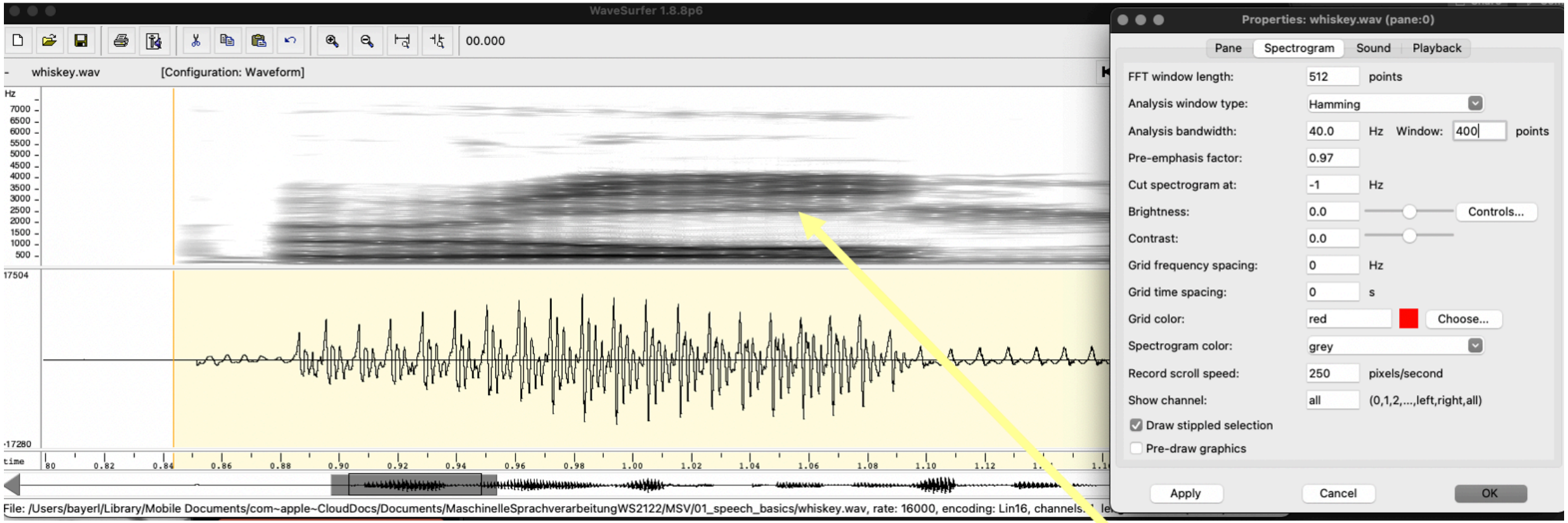
Wideband Speech Spectrogramm



Horizontal bands match formant frequencies

Vertical lines/ structure implies sudden closure of the vocal folds

Narrowband Speech Spectrogram

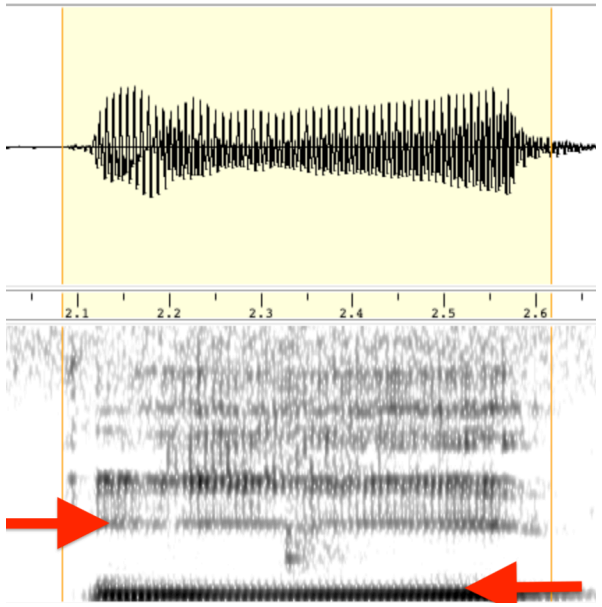


Harmonics have higher

energy at formant resonances

Spectrogram

- **Time** on the x-axis
- **Frequency** on the y-axis
- **Amplitude** represented by color / intensity
- **Spectrogram** is underspecified unless the window length is given



Trade-off between time and frequency resolution

Wideband Speech Spectrogramm

Uses a short time window, which provides:

- good time resolution
- poor frequency resolution

Narrowband Speech Spectrogramm

Uses a longer time window, which provides:

- poor time resolution
- good frequency resolution

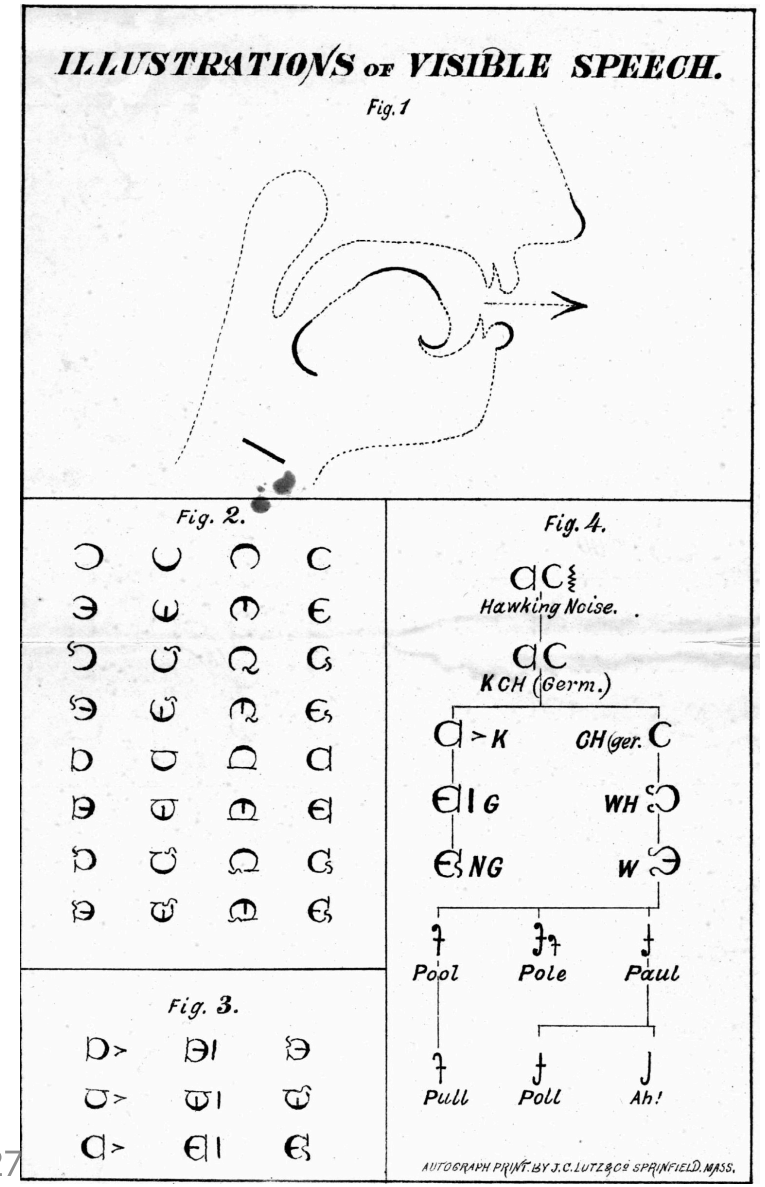
Writing Systems

Writing Systems

- writing is an ancient technology
  - invented as a **communication aid**
  - **! not all languages have a writing system !**
- Writing allows information to be:
  - **stored**
  - **retrieved**
  - **transmitted**
- the basic unit is the **word** (because it caries meaning)
- meaningful **utterances** are made up from sequences of words
- Languages deceit words using **alphabets**, **syllabaries**, **logographs** or **ideographs**
- Writing systems that use alphabets and syllabaries are **"phonetic"**, i.e., based on how the words sound
- some languages have a closer relationship between the spoken and written forms as others (e.g., Korean vs French)

Sound-Based Writing Systems

- Sound-based writing can be thought of as a kind of visible speech
- "Visible Speech"
  - sound-based writing system
  - invented in 1867 by Alexander Melville Bell (father of Alexander Graham Bell)
- Visible Speech was intended to help hearing-impaired people learn spoken language
- The symbols were representations of the positions of the vocal organs
  - hence it was independent of a particular language or dialect

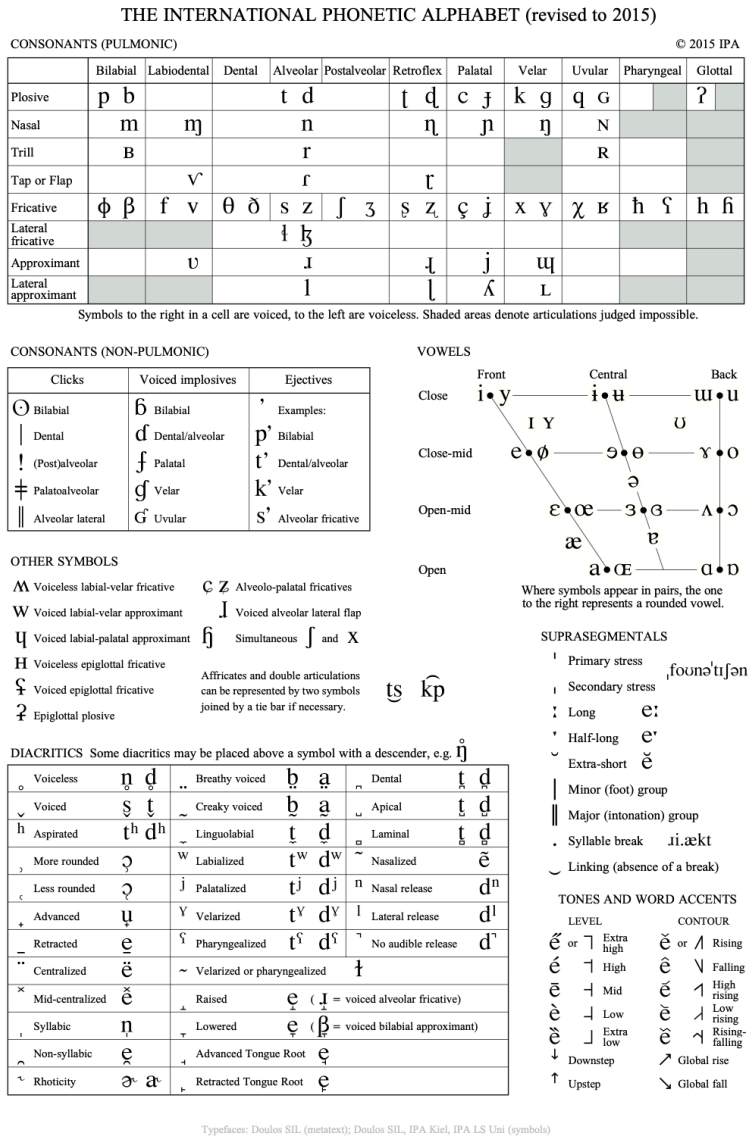


Phonetic Transcription

- Languages select from 200-300 basic speech sounds known as **phones**
- The International Phonetic Association (IPA), founded 1886, has agreed on a set of standard symbols for representing any speech sound
- A trained **phonetician** can use the IPA symbols to transcribe spoken utterances from any language (even a newly discovered one) or any speaker
- The symbols in an IPA phonetic transcription are written between square brackets, e.g. ...
  "hello" → [hələʊ]
  "speech processing" → [spiːtʃ pɹəʊsɛsɪŋ]

The International Phonetic Alphabet (IPA)

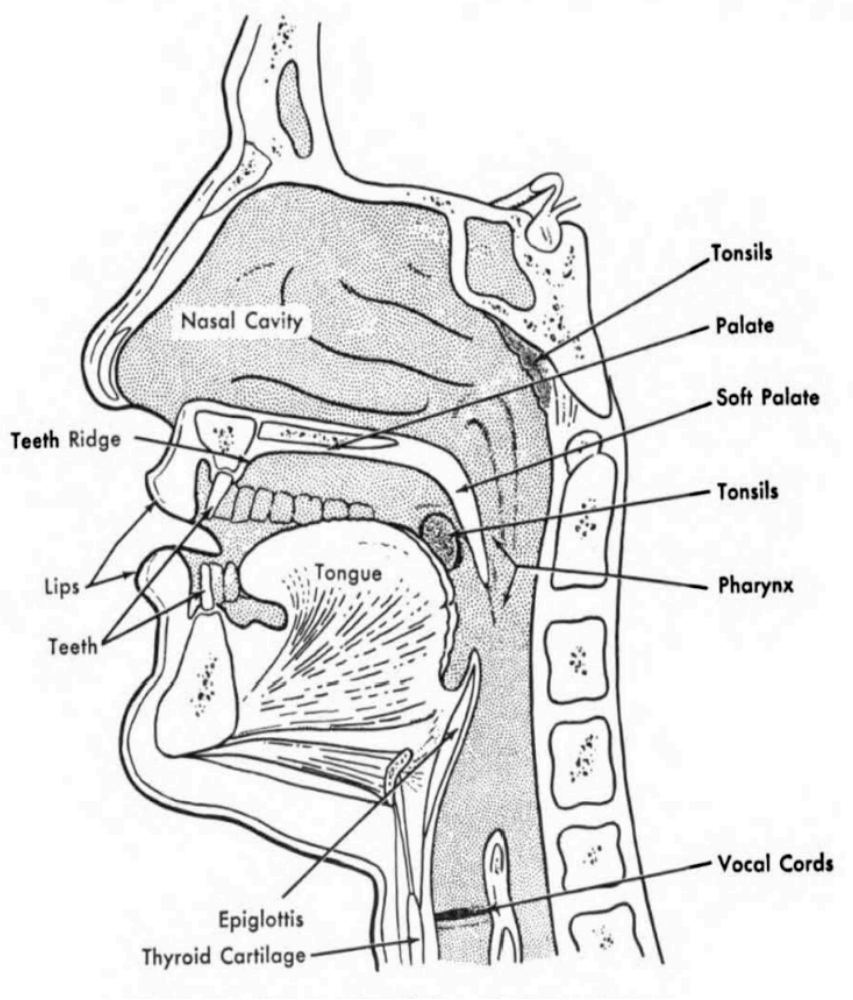
Check out for comparison and sample recordings of German and Spanish sounds:
https://www.ipds.uni-kiel.de/kjk/pub_exx/kk2009_4/cons_spa-ger/cons_spa-ger.html



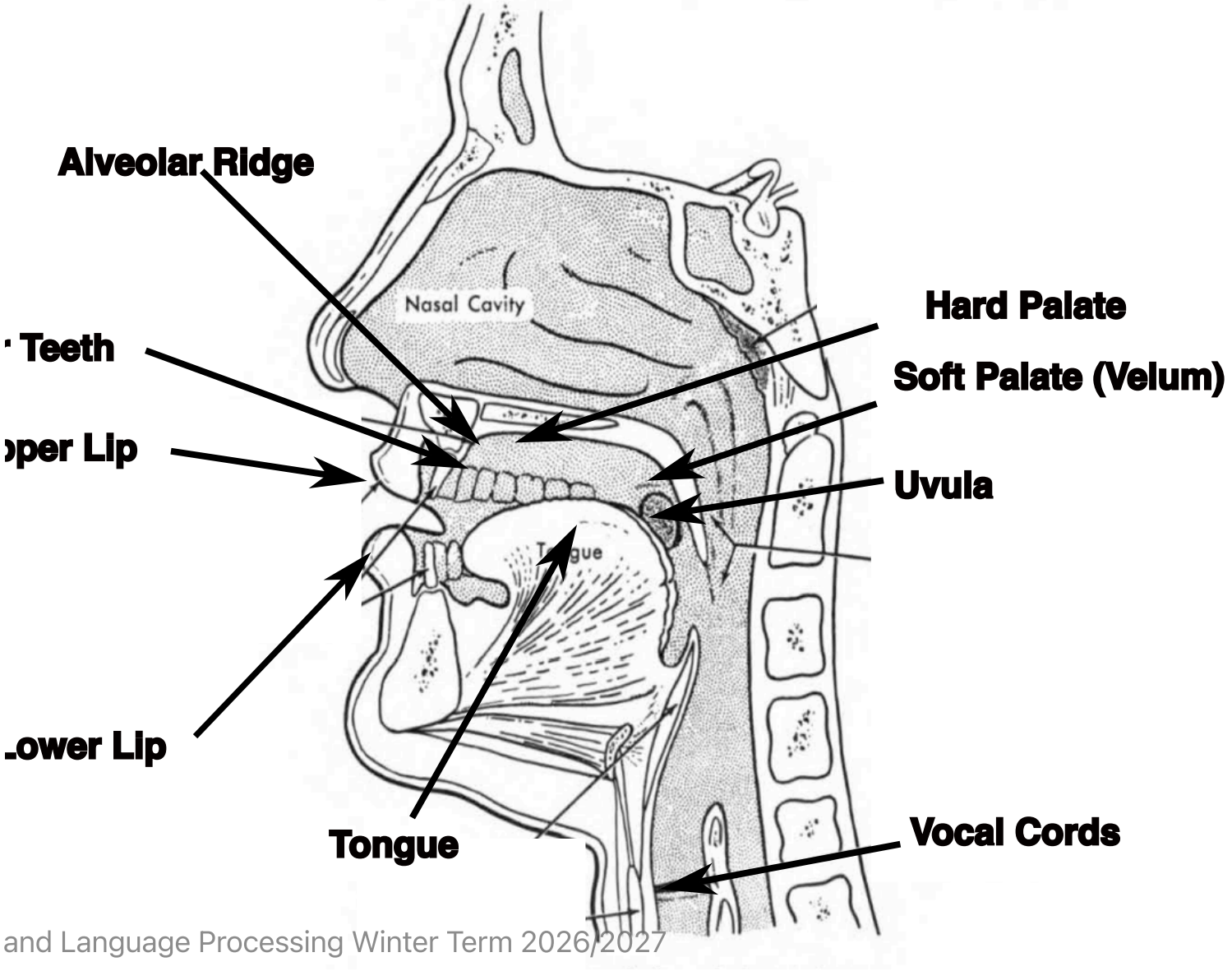
How sounds are formed

The Resonant Cavities

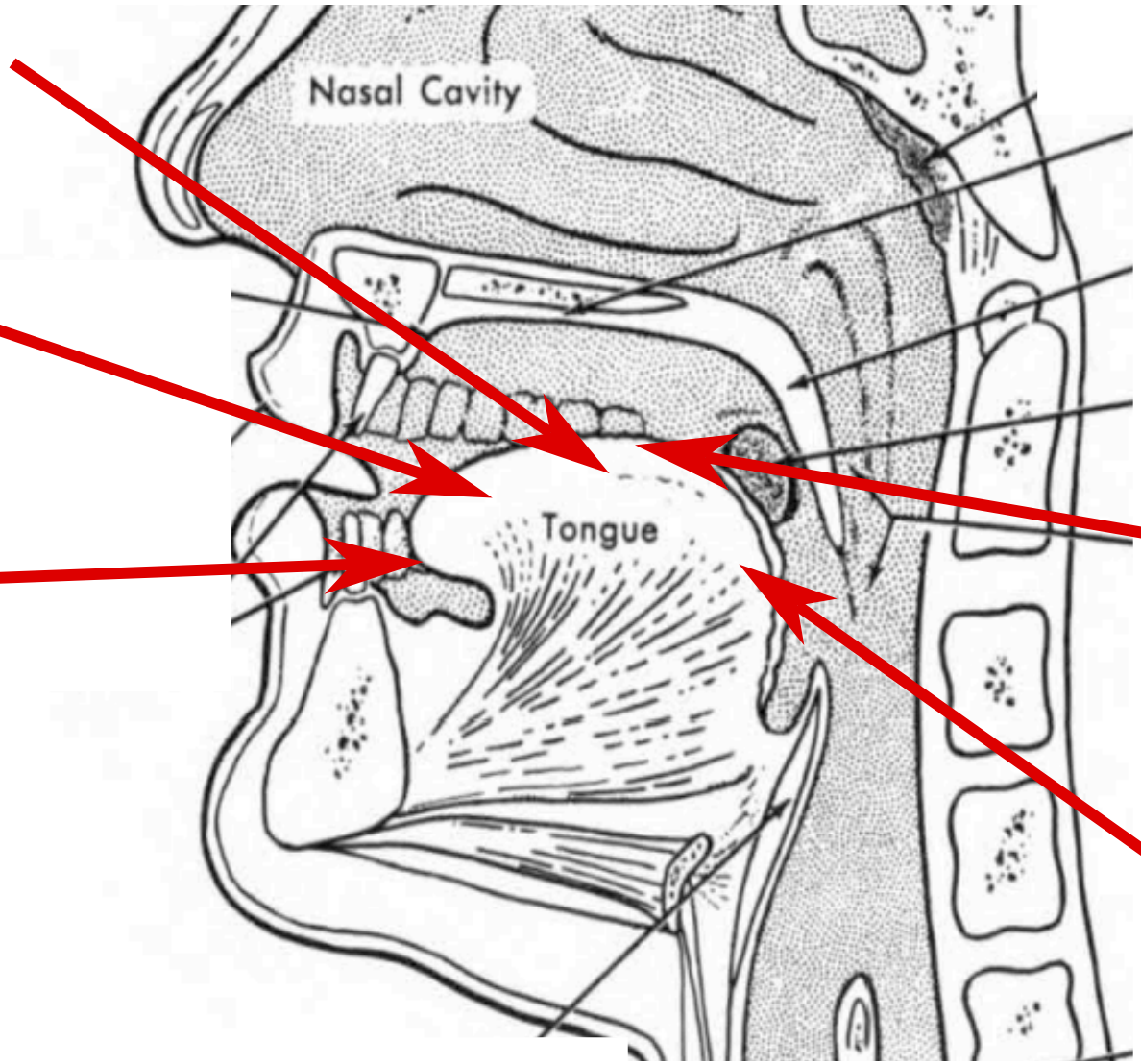
- Nasal Cavity
- Oral Cavity
- Pharyingial Cavity
- Laryngial Cavity



The "Articulators"



The Tongue


Tongue Front
Tongue Blade
Tongue Tip
Nasal Cavity
Tongue
Tongue Back
Tongue Root

The Syllable

- Syllables are the basic building blocks of spoken language
- Shortest stretch of speech

Syllable Structure:

- Syllables consist of:
  - **vowels**: sound segments produced using an unobstructed configuration of the vocal tract
  - **consonants**: sound segments produced where the airflow is at least partly obstructed
- A simple "CVC" syllable consists of a consonant, a vowel, and a consonant
- which corresponds to the opening and closing of the mouth
- **Nucleus**: the most sonorous part of a syllable
- **Onset**: the segments that come before the nucleus
- **Coda**: if there are any segments after the nucleus they’re called the coda

Words can be:

- **monosyllabic** (one syllable)
- **polysyllabic** (more than one syllable)

Speech Sounds

In articulatory phonetics, speech sounds are classified according to:

- where the air stream comes from
- whether air is going in or out
- whether the vocal cords are vibrating
- the location of any constriction
- how the sound is made
- the position of the tongue
- the shape of the lips

Consonants

Voice

- degree of voicing (voiced-voiceless)
- voice quality (e.g., modal, creaky, falsetto, breathy, etc.)

Place

- the physical location of the constriction in the vocal tract

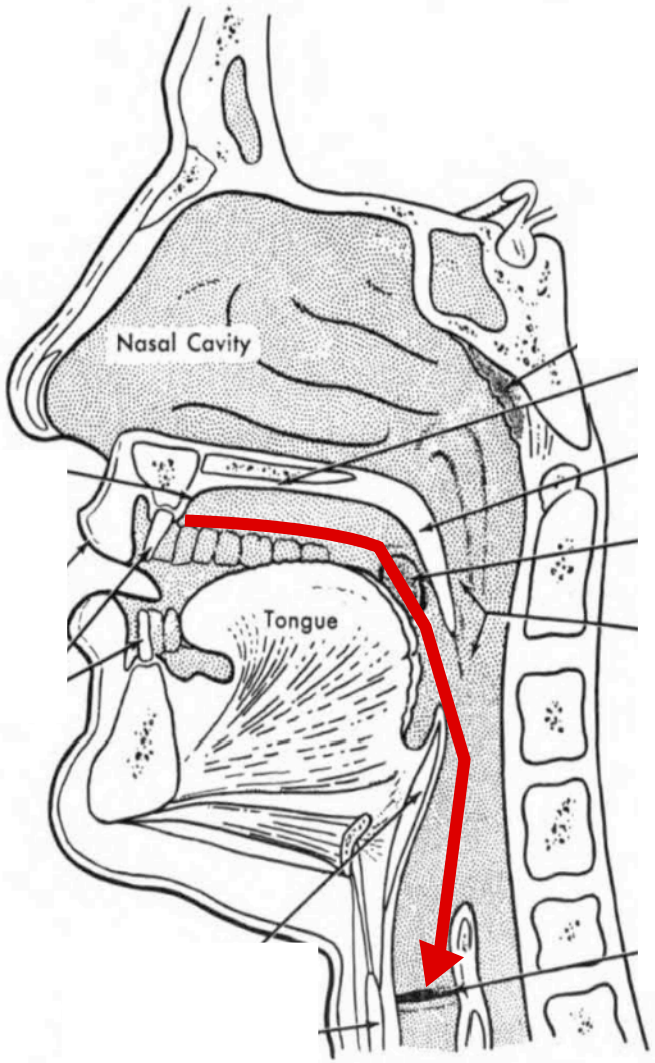
Manner

- the way in which the airstream is modified by the primary and secondary articulators
- the degree of stricture (closure, narrowing or approximant)

The Place of Articulation

Top to Bottom:

- Bilabial
- Labiodental
- Dental
- Alveolar
- Postalveolar
- Retroflex
- Palatal
- Velar
- Uvular
- Pharyngeal
- Glottal



Place of Articulation

Place of Articulation	Active Articulator	Passive Articulator	Example Sounds
Bilabial	upper and lower lips	none	[p b m]
Labiodental	lower lip	upper front teeth	[f v]
Dental	tongue tip	upper front teeth	[θ ð]
Alveolar	tongue tip or blade	alveolar ridge	[t d n l s z]
Postalveolar	tongue tip or blade	rear of alveolar ridge	[ɹ ʃ]
Retroflex	tongue tip	hard palate	[ʈ ɖ ɳ]
Palatal	tongue front	hard palate	[j ɲ]
Velar	tongue back	soft palate	[k g ŋ]
Uvular	tongue back	uvula	[q ɢ]
Pharyngeal	tongue root	rear wall of pharynx	[ħ]
Glottal	vocal folds	none	[h ʔ]

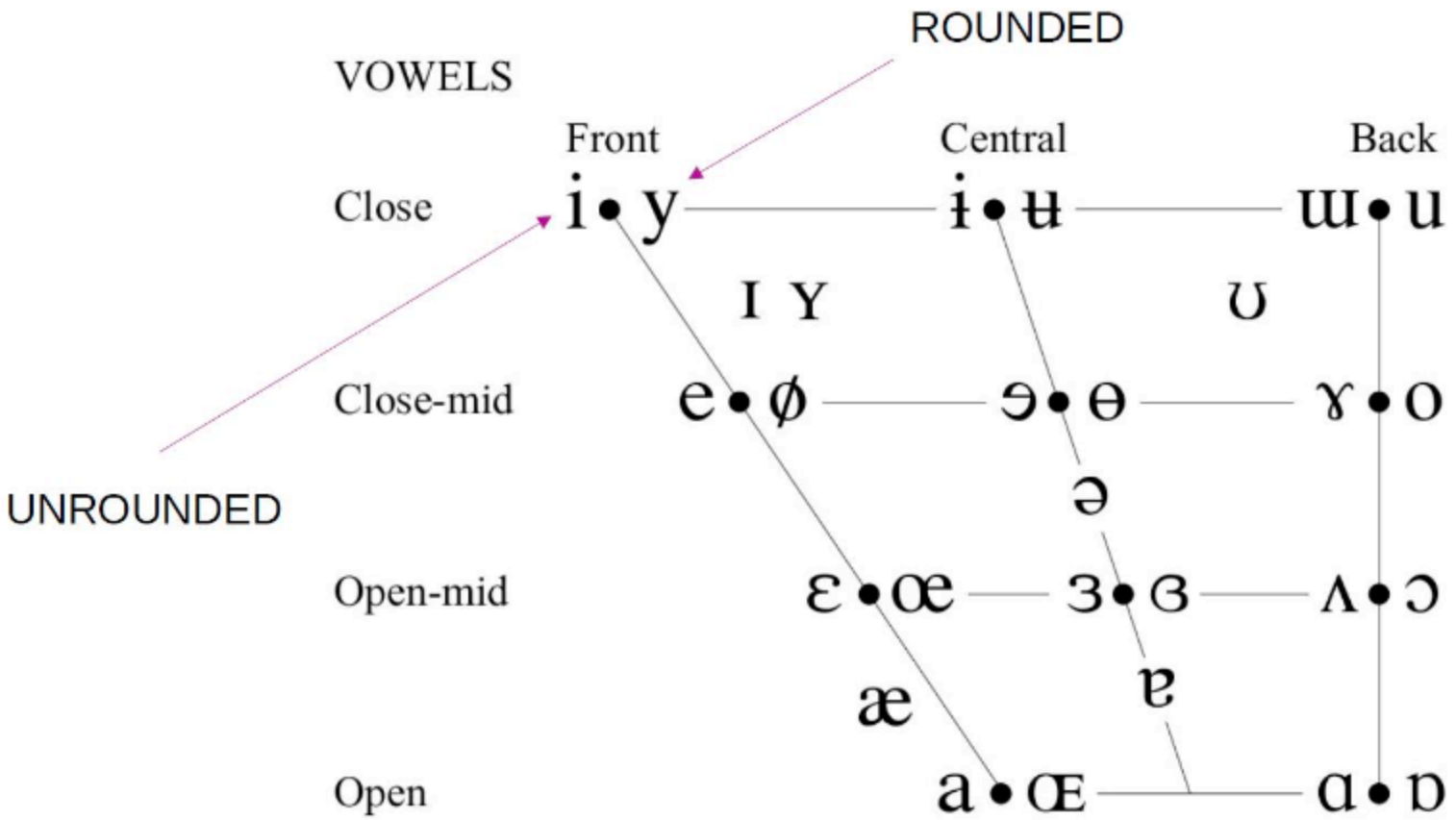
Manner of Articulation

- **Stops**
  - complete blockage of the airstream
  - can be produced at many different places of articulation
  - e.g. **plosives**[p t k b d g], **affricates** [tʃ dʒ],**nasals** [n m]
- **Fricatives**
- sound is produced with a very narrow opening
- the resulting airflow is turbulent and hence noisy – e.g. [fsh]
- **Approximants**
  - stricture is not narrow enough to cause turbulence – e.g. lateral[l], glides[wʊj]
- **Taps & Trills**
  - a single closure much shorter than for a plosive,e.g.[ɾ] – a series of rapid tap-like closures, e.g.[r]

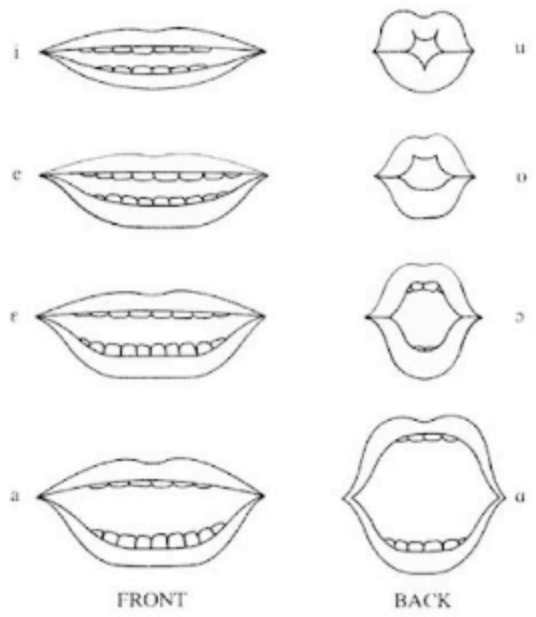
Vowels

- Vowels are articulated by:
  - raising the front or the back of the tongue owards the roof of the oral cavity
  - shaping the lips
  - thereby changing the "vowel quality"
- Vowel quality is governed by:
  - "vowel height": high / low
  - "vowel location": front / back
  - "lip position": rounded / unrounded
- The mid-central vowel [ə] is called "schwa"
- ! Any vowel can be a "schwa" (e.g., sal**a**d, el**e**phant, an**i**mal, lem**o**n, **u**pon)

IPA Vowels

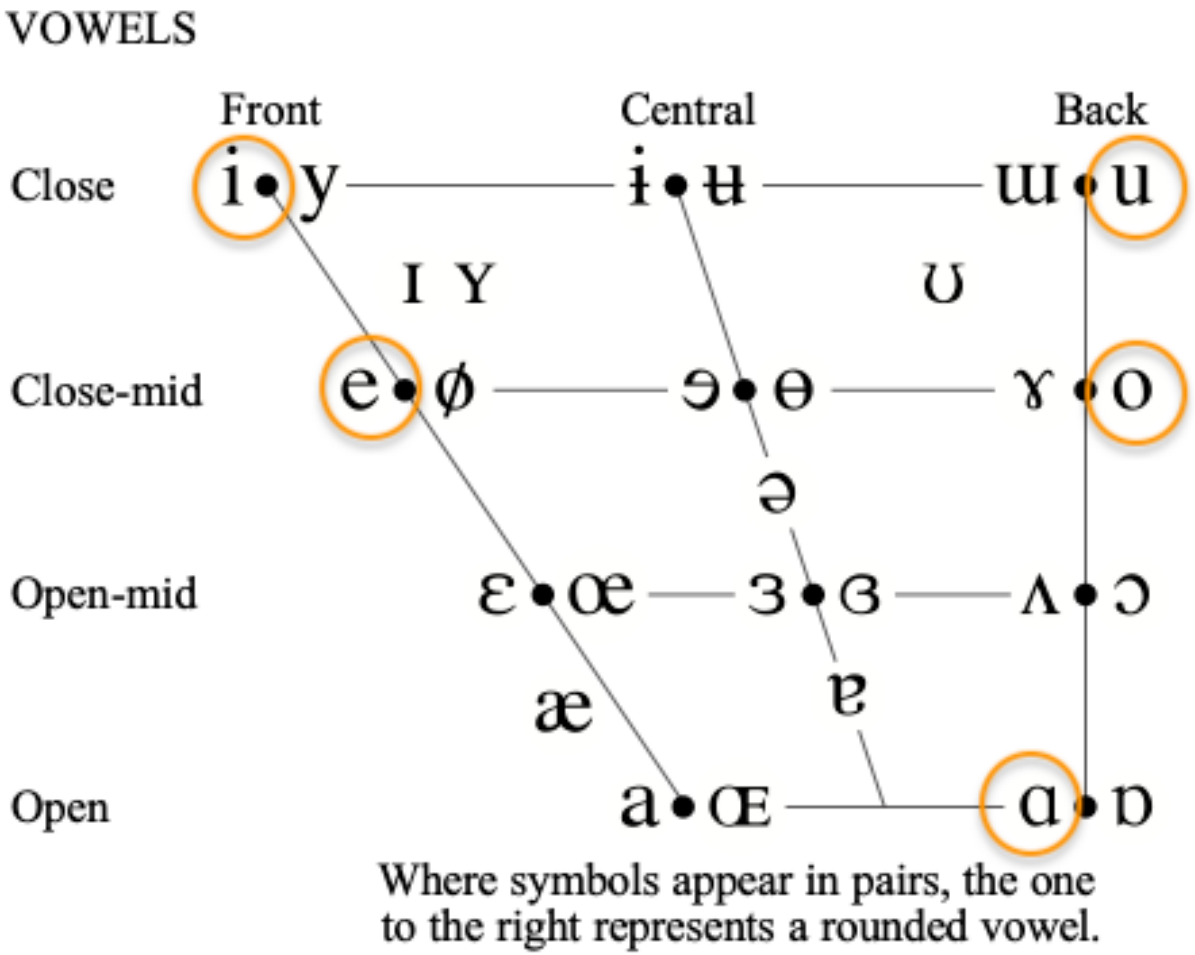


Where symbols appear in pairs, the one to the right represents a rounded vowel.



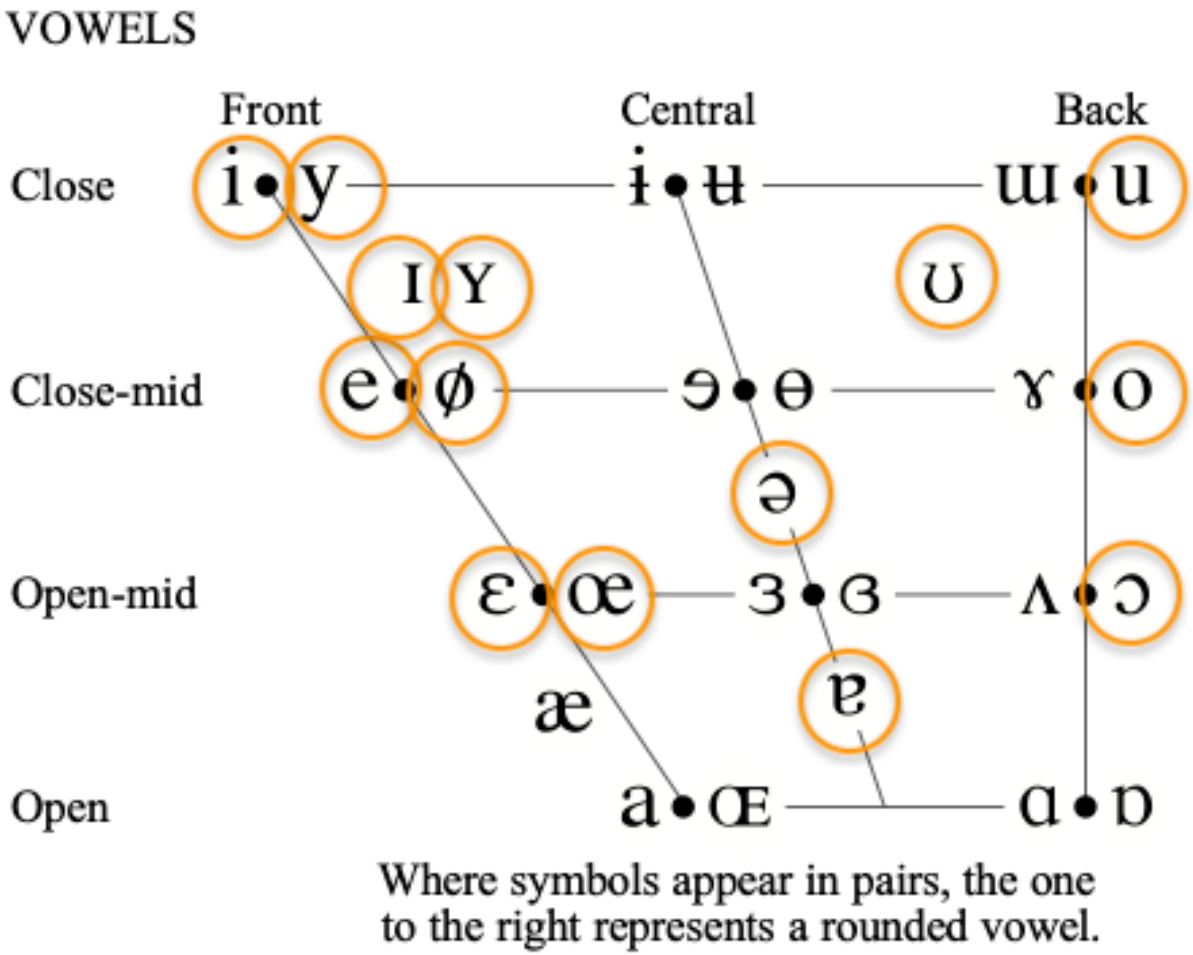
IPA Vowels

Spanish Vowels



IPA Vowels

German Vowels



IPA Pulmonic Consonants

CONSONANTS (PULMONIC) © 2015 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

IPA Pulmonic Consonants

CONSONANTS (PULMONIC) © 2015 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

Stationary Sounds

- Some speech sounds("continuants") can be sustained over time and retain their phonetic quality

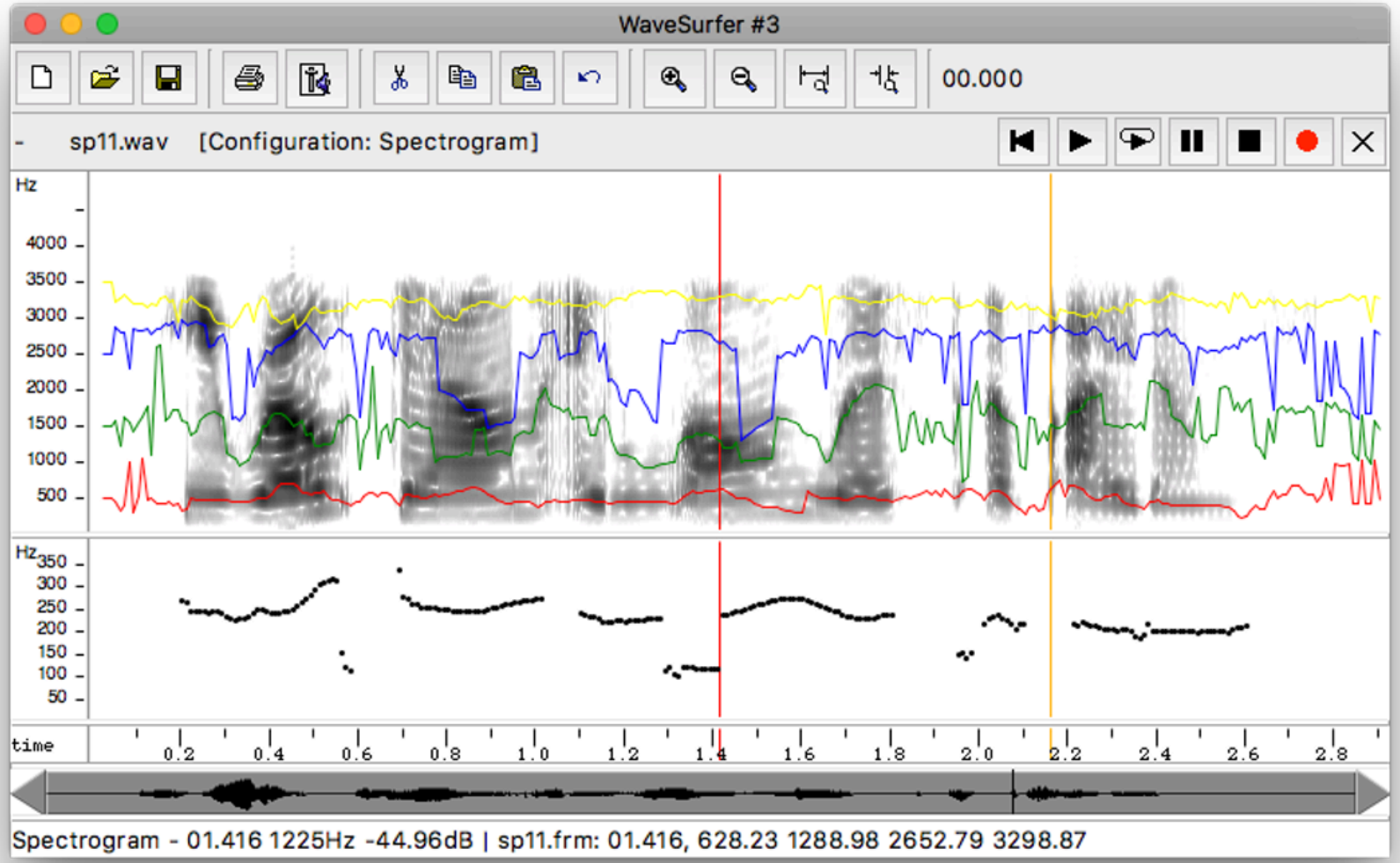
For example:

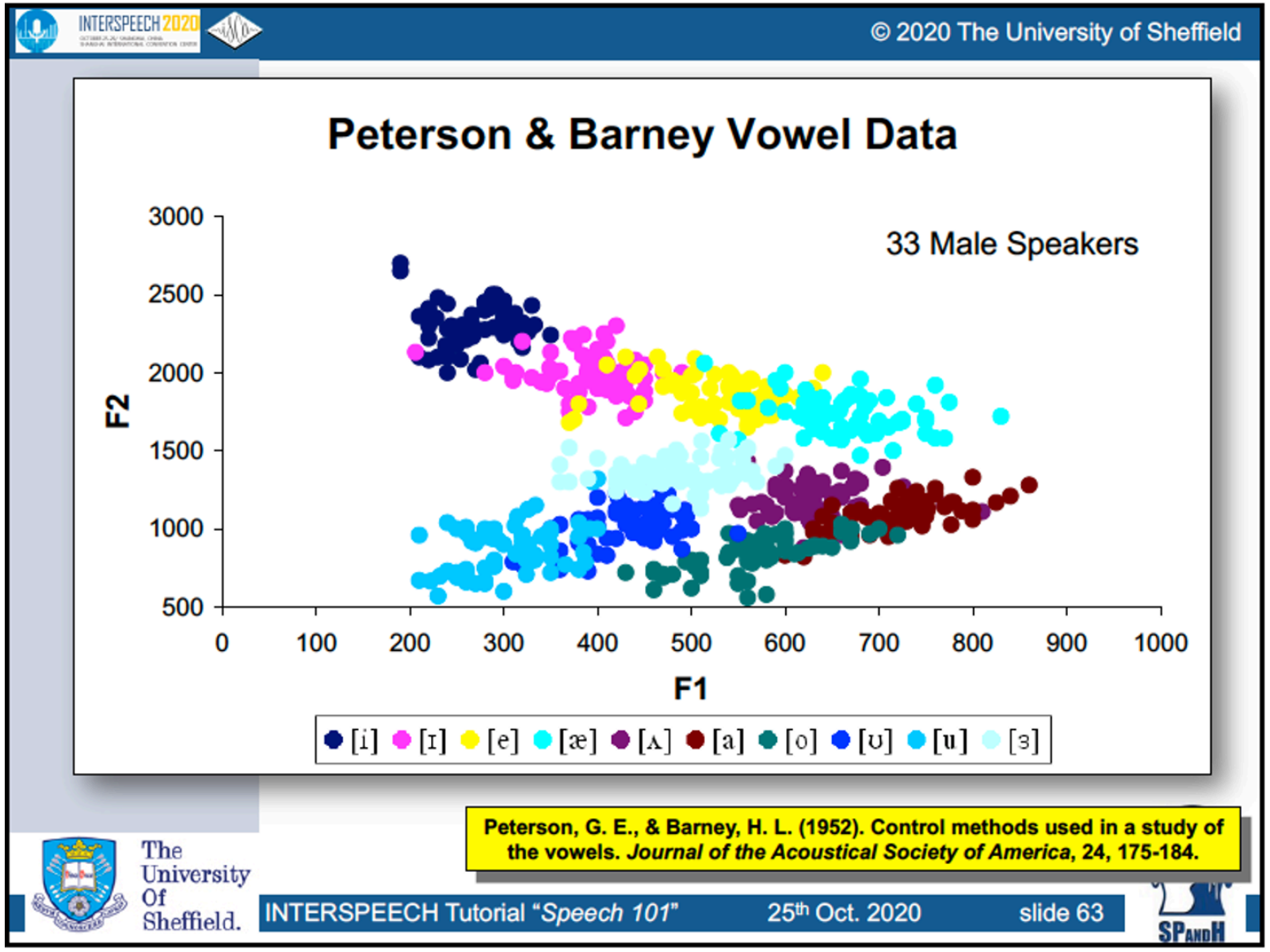
- monophthong vowels, e.g. [iː æ uː] – all fricatives, e.g. [f θ v ð s ʃ z ʒ]
- some approximants, e.g. [ɹ l]
- all nasals, e.g. [n m ŋ]

Formants

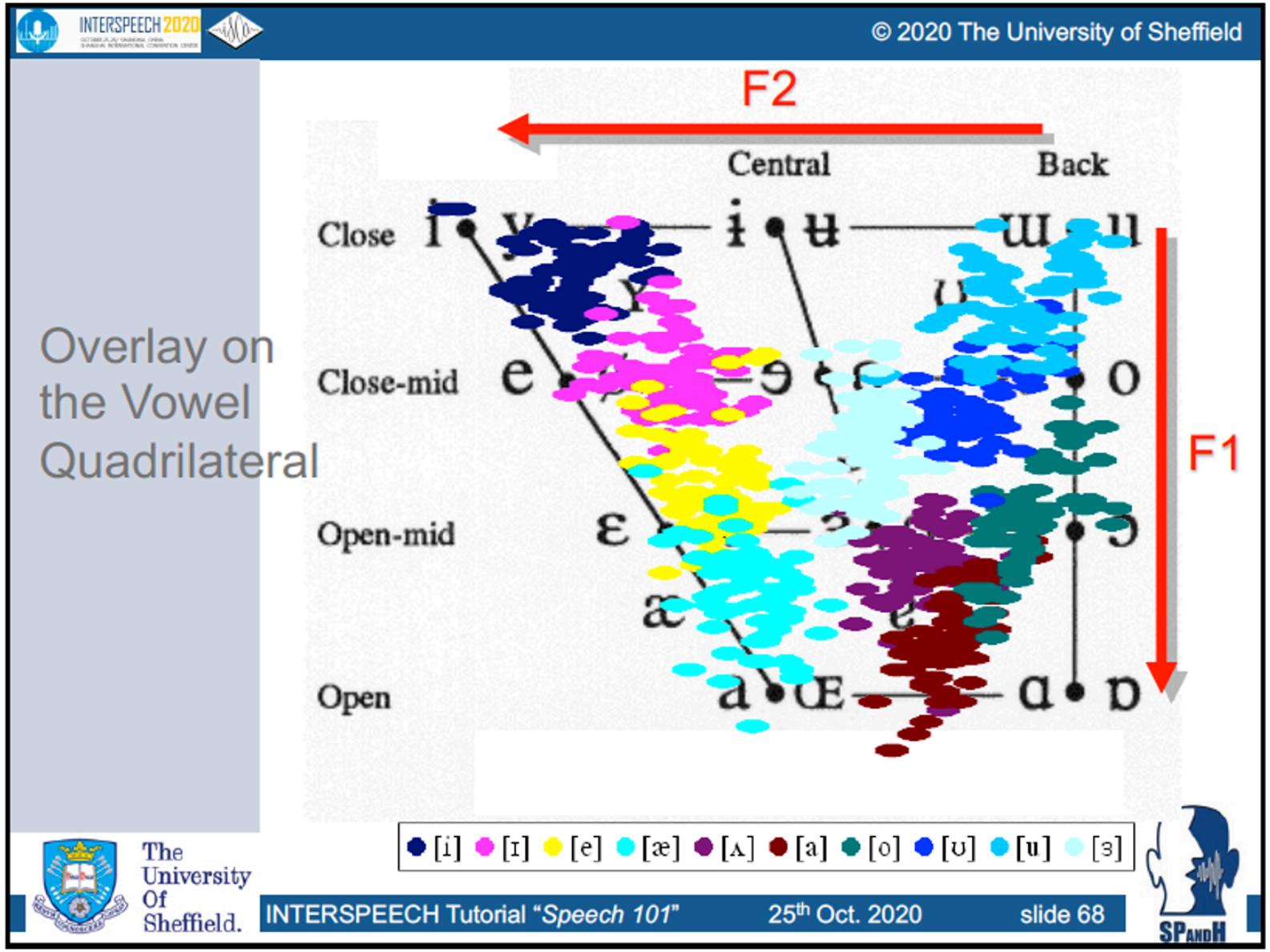
- a formant is the broad spectral maximum
- as a result of acoustic resonance of the human vocal tract
- $\neq f_0$ (fundamental frequency) which is the rate of vocal fold vibration, i.e., the production mechanism
- the first two formants are important in determining the quality of vowels
- Vowels will almost always have four or more distinguishable formants (up to six)

Formants

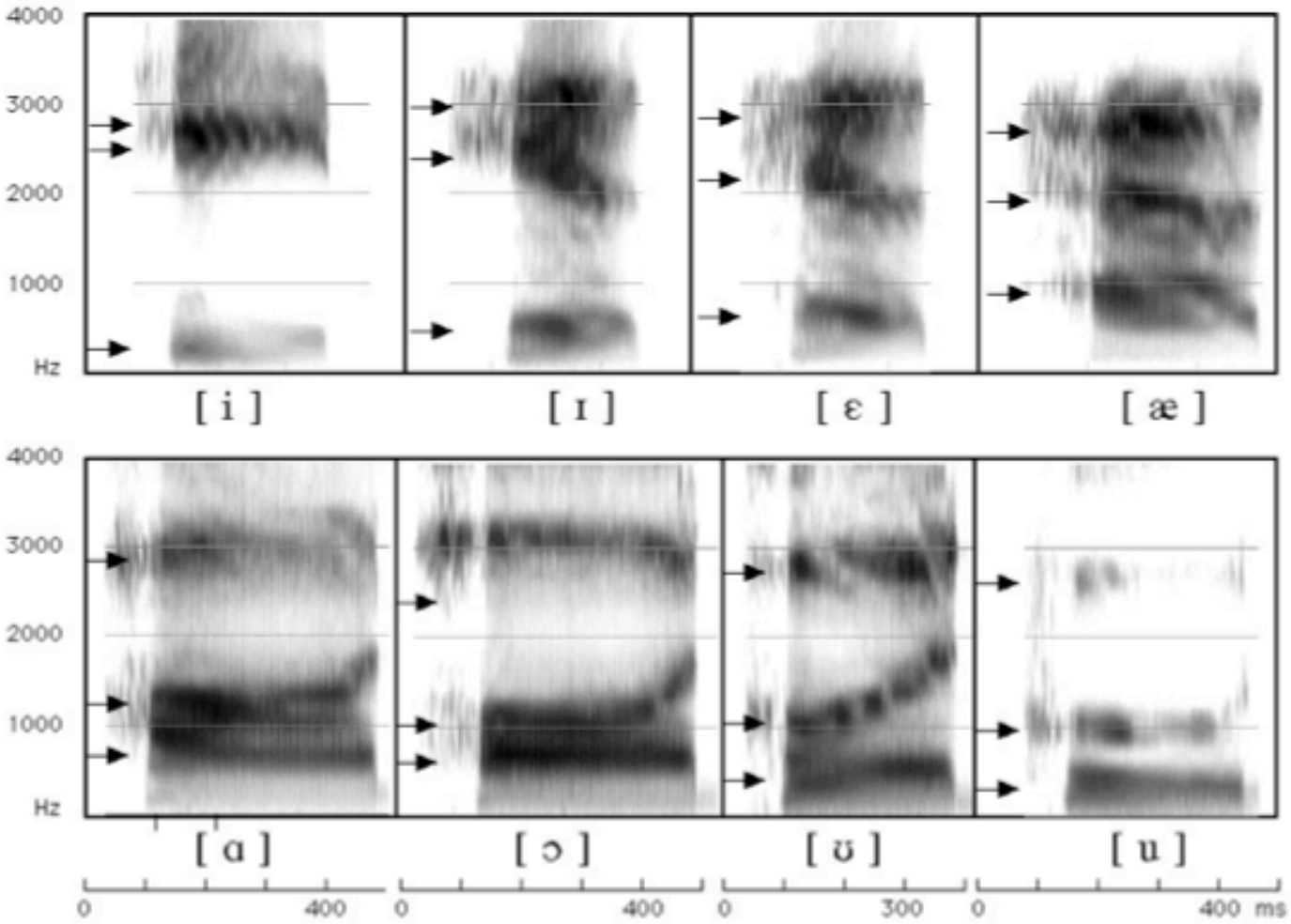

WaveSurfer #3
00.000
- sp11.wav [Configuration: Spectrogram]
Hz
4000
3500
3000
2500
2000
1500
1000
500
Hz
350
300
250
200
150
100
50
time
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
2.6
2.8
Spectrogram - 01.416 1225Hz -44.96dB | sp11.frm: 01.416, 628.23 1288.98 2652.79 3298.87


INTERSPEECH 2020
© 2020 The University of Sheffield
Peterson & Barney Vowel Data
33 Male Speakers
F2
F1
3000
2500
2000
1500
1000
500
0
100
200
300
400
500
600
700
800
900
1000
[i] [ɪ] [e] [æ] [ʌ] [a] [o] [ʊ] [u] [ɜ]
Peterson, G. E., & Barney, H. L. (1952). Control methods used in a study of the vowels. Journal of the Acoustical Society of America, 24, 175-184.
The University Of Sheffield.
INTERSPEECH Tutorial "Speech 101"
25th Oct. 2020
slide 63
SPandH

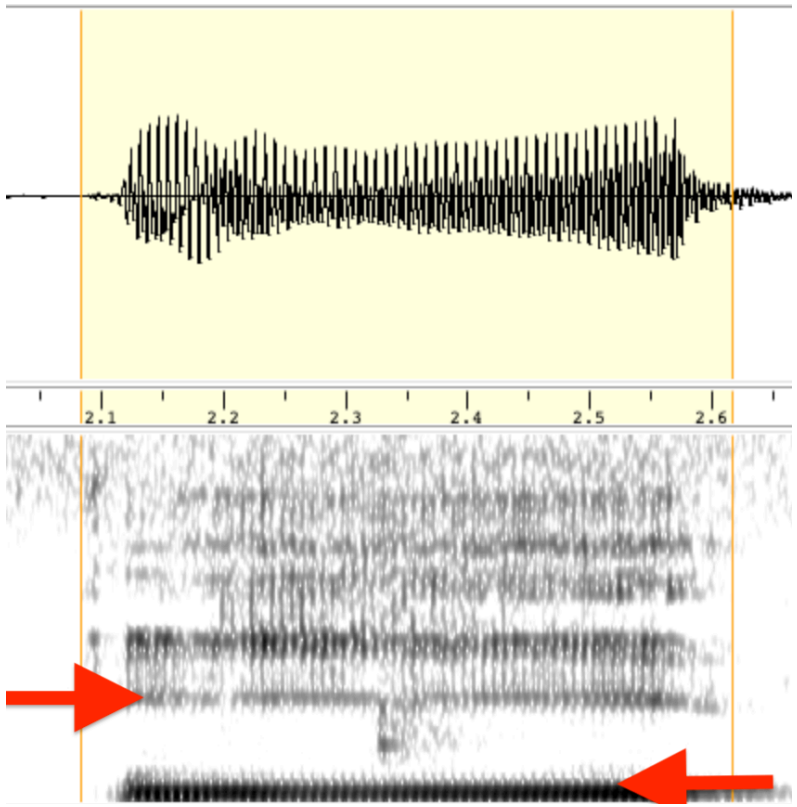
Overlay on the Vowel Quadrilateral

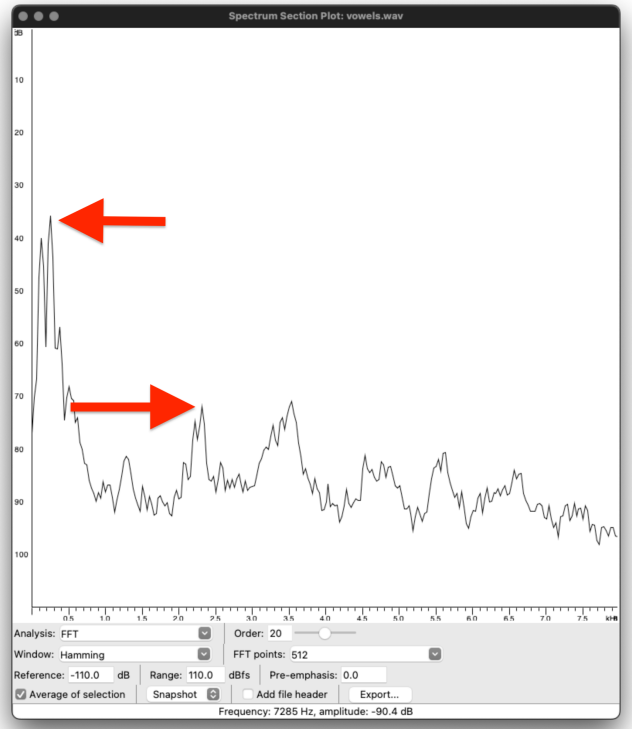


Formants Spectral Analysis of Vowels

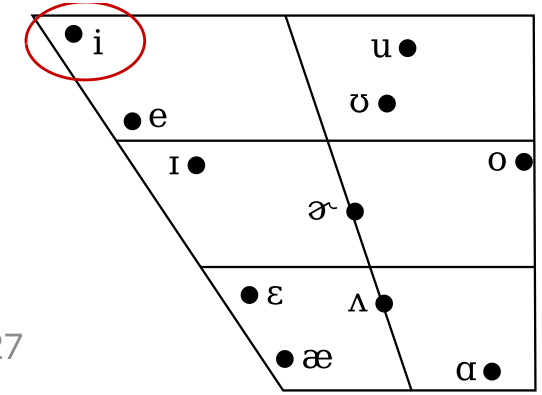


Formant Analysis





- "i"
- **Low F1** (~200Hz- 300Hz)
- **High** F2 (~2100Hz)



Non-Stationary Sounds

- The essence of some speech sounds is that they change over time

For example:

- diphthongs, e.g. [eɪ əʊ]
- plosives, e.g. [p b t d k g]
- affricates, e.g. [tʃ dʒ]
- glides, e.g. [w j]

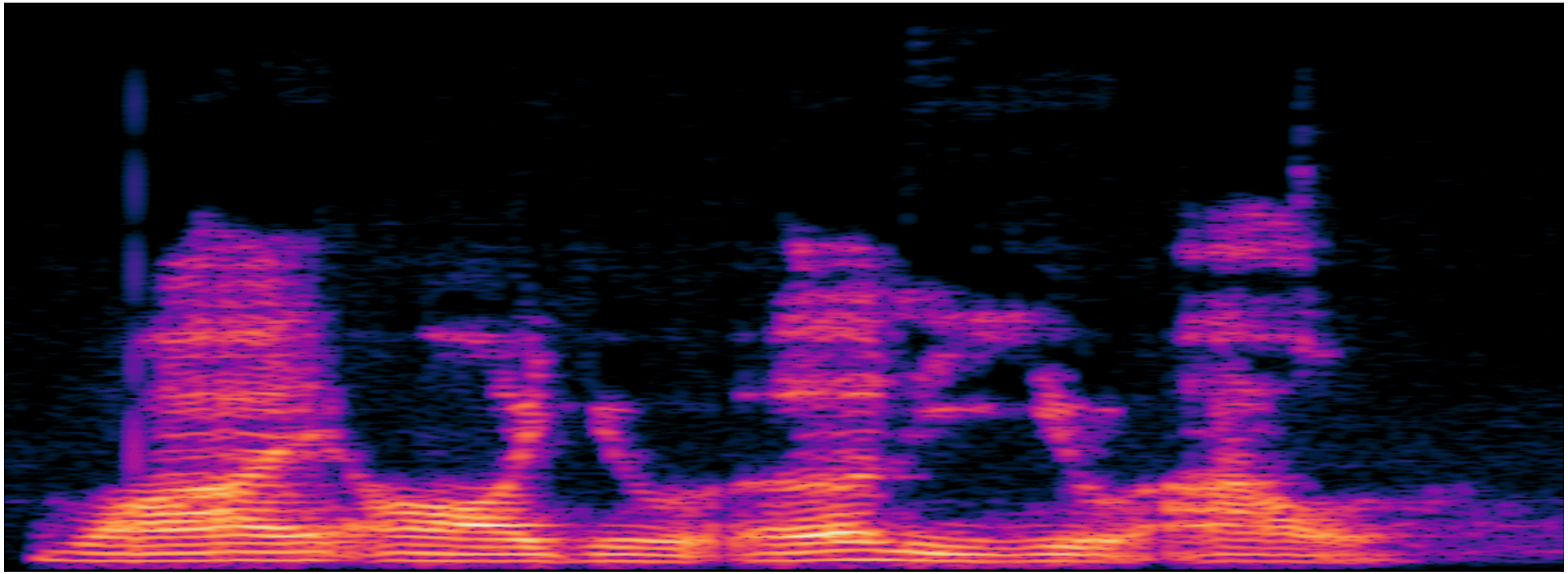
Coarticulation

Coarticulation is the phenomenon in which the articulation of one sound influences the articulation of another sound

- when working with speech (as an engineer), this is a form of **context dependency**
- to a phonetician, it is a consequence of **motor planning**
- coarticulation can be both short and long-range
  - e.g., the influence of nasalization or lip rounding can span several syllables
  - not limited to adjacent phones

Coarticulation

"Why are you early you owl?"



Prosody

Prosody

Prosody is one way in which speech can be made more expressive.
It allows speakers to signal **important** information in what they say.

It allows **different** words to be **emphasized** and is **criticial** to communicating **un**ambiguous **meanings**.

It's a **bit** like **punctuation** in written language, but prosody carries **much** more information **!**

Prosody

- prosodic features span:
  - several speech segments
  - several syllables
  - whole utterances
- such suprasegmental behavior includes...
  - lexical stress
  - lexical tone
  - rhythmic stress
  - intonation

Lexical Stress:

- prominence of syllables within words
- stressed syllables are usually (and/or) longer, louder, and higher in pitch

Lexical Tone:

- ≈ half the world's languages use the pitch pattern to distinguish between one word and another
  • Modern Standard Chinese has four lexical tones: high level, high rising, low falling-rising, high falling

Intonation

- refers to pitch variation that doesn't affect the meaning of the words, but does affect the meaning of an utterance
- e.g., "It's cold!" vs. "It's cold?"
- Intonational Accent: contrasting pitch movement on a syllable
  - e.g., "This is **my** book" vs. "This is my **book**"

Speech is...

continious

- speech is a continious signal
- it is not a sequence of discrete sounds

variable

- 3000-8000 languages, (6500 "living")
- 83% are limted to a single country
- 52% are spoken by less than 10000 people
- 28% are spoken by less than 1000 people
- by 2100, around half of these languages will be extinct
- everyone speaks differently / has their own style
- dialects

Speech is...

adaptive

- to the listner
- to the task
- to the cognitive load
- to the environment (Lombardi Effect, louder + hyperaticulation)

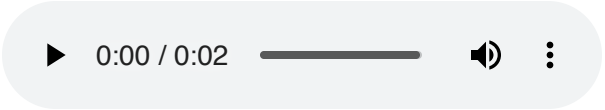
efficient

- "fish and chips" $\rightarrow$ "fish'n'chips"
- "bread and butter" $\rightarrow$ "brem'm butter"

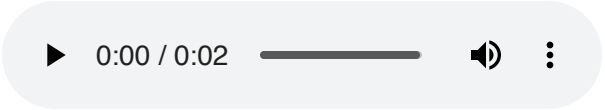
ambigious

- "to" vs "two" vs "too"
- "hear" vs "here"
- "This nudist play will wreck a nice beach" vs "This new display will recognise speech"

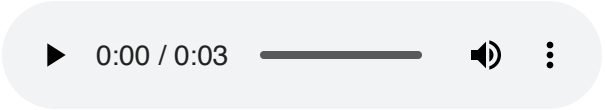
- It's not hard to recognize speech



- It's not hard to wreck a nice beach



- Bonus



Speech is...

multimodal

- Speech is not just an acoustic signa
- listeners can derive large benefit from seeing the speaker's lips
- "MCGurk effect"

alle profs und lehrer sagen random ja

Speech is...

Disfluent

- filled pauses
- speech disorders

More than language

Rich in "extra-linguistic" information

- breathing noises
- lip-smacks
- Rich in "para-linguistic" information
- individuality
  – personality
  – attitude
  – emotion

Noisy

0:00 / 0:02

0:00 / 0:02

0:00 / 0:02

0:00 / 0:02

0:00 / 0:02

0:00 / 0:02

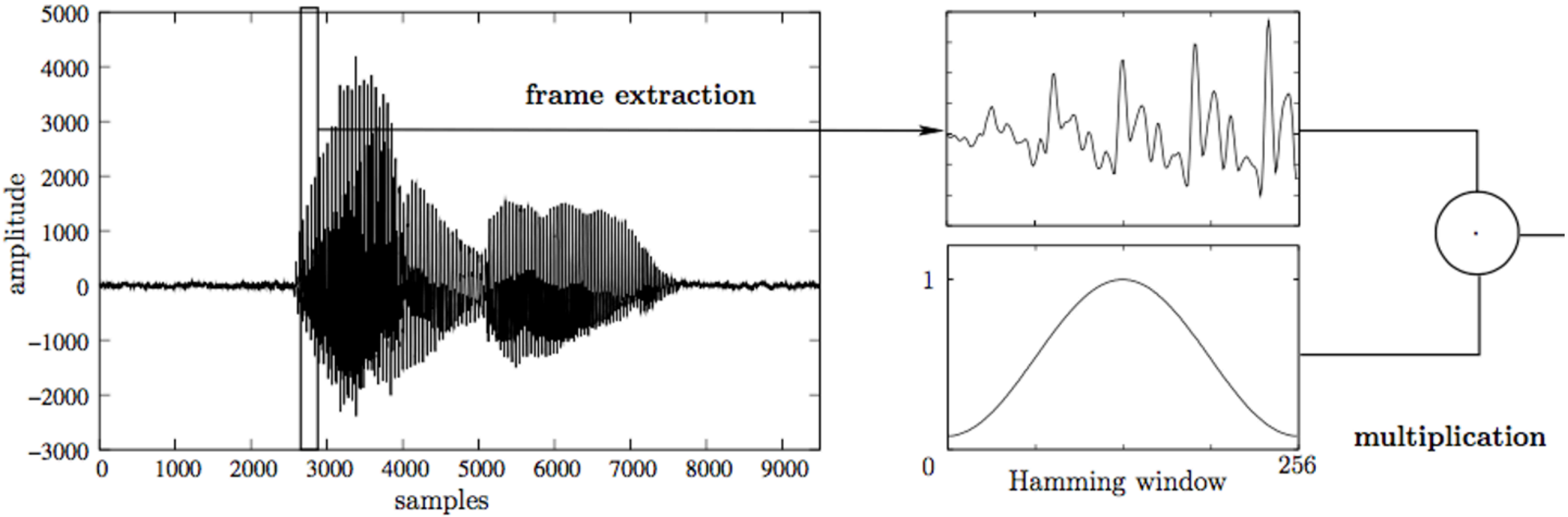
Using (Speech) Audio with AI Systems

Problem: getting a continuous signal into a computer

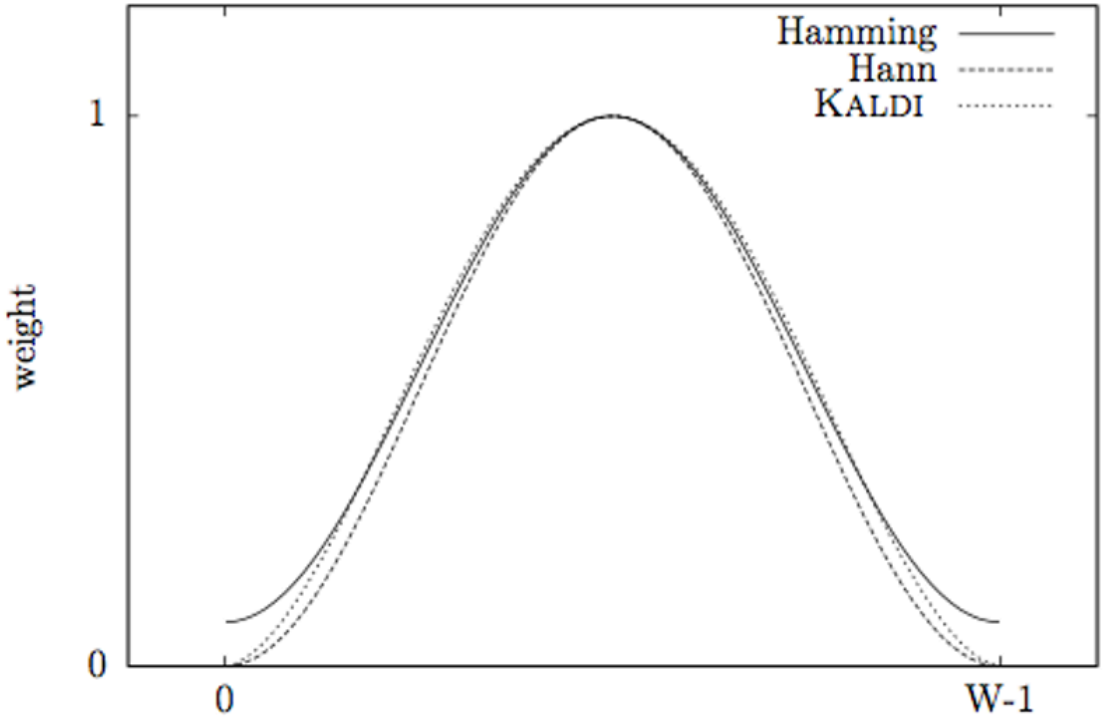
i.e., getting features from the signal

Two main paradigms: Signal Processing and Machine Learning based features

Short-time analysis of speech signals



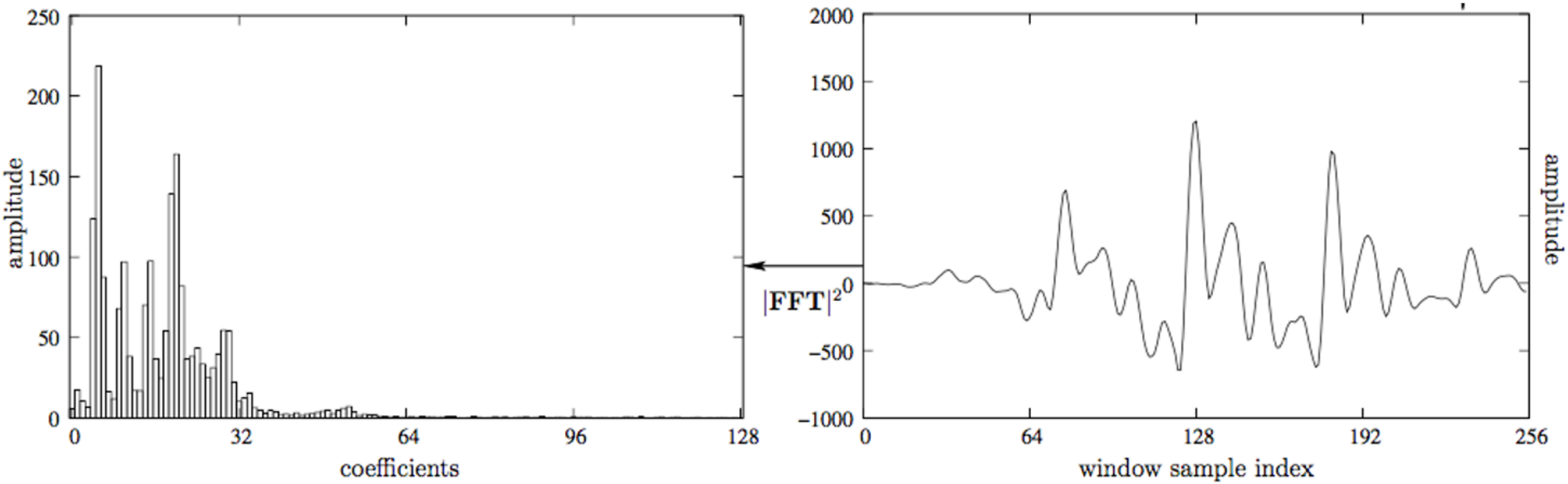
Windowing ins short-time analysis



name	*weighting function*
Hamming	$\text{hamming}(n) = 0.54 - 0.46\cos\left(\frac{2n\pi}{W-1}\right)$
Hann	$\text{hann}(n) = 0.50 - 0.50\cos\left(\frac{2n\pi}{W-1}\right)$
KALDI	$\text{kaldi}(n) = \left(0.50 - 0.50\cos\left(\frac{2n\pi}{W-1}\right)\right)^{0.85}$

Spectral Analysis

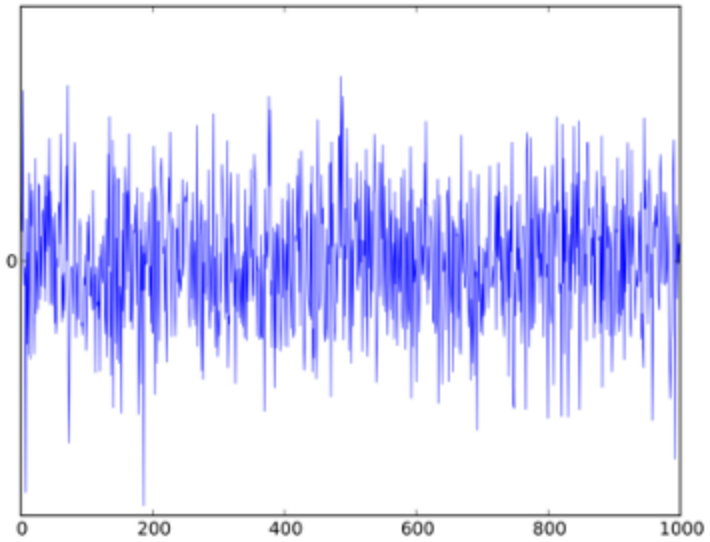
Fourier Transform of the windowed signal



Fourier Series

$$f(t) = a_0 + \sum_k^\infty a_k \cos(n2\pi kt) + b_k \sin(2\pi kt)$$

- periodic function $f(t)$
- sum of sinusoidal functions $\sum_k^\infty a_k \cos(n2\pi kt) + b_k \sin(2\pi kt)$
- frequency k



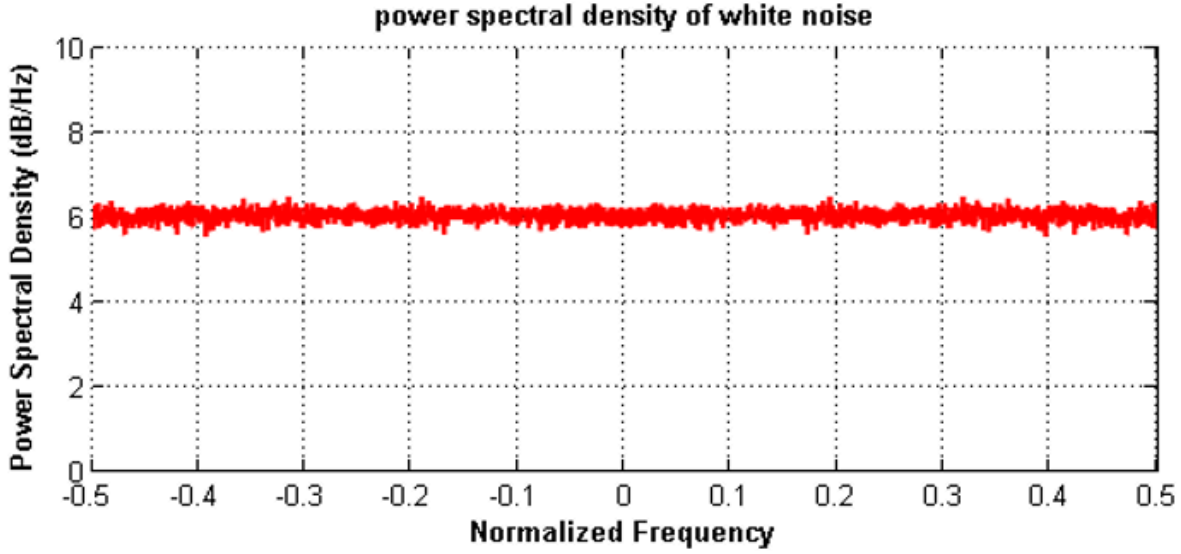


Image source: www.gaussianwaves.com%2F2013%2F11%2Fsimulation-and-analysis-of-white-noise-in-matlab

Image source: https://en.wikipedia.org//White_noise#/media/File:White_noise.svgwiki

Fourier Transform

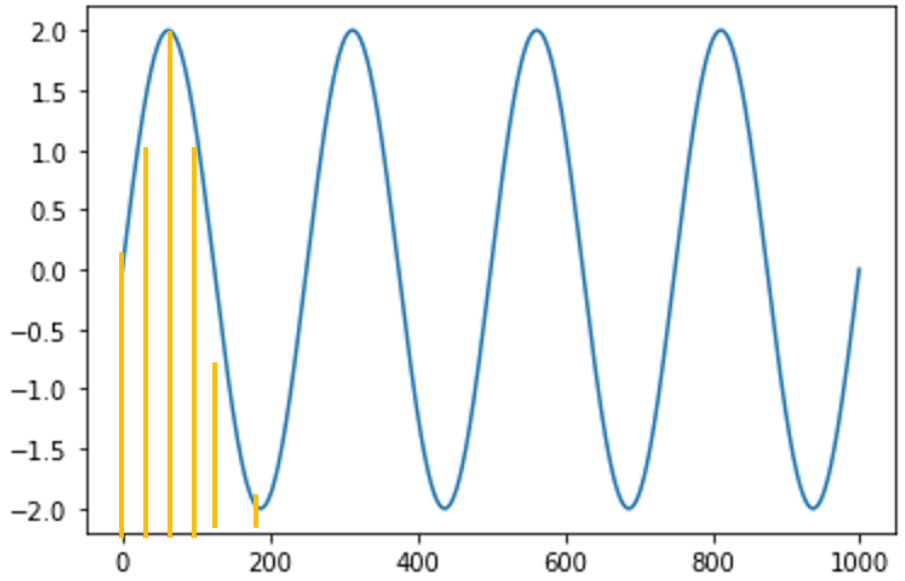
- examines the similarity of a signal to sinusoidal functions at different frequencies
- using an analysis function (sine and cosine) to determine the amplitude and phase of the signal at each frequency
- similarity to correlation analysis

$$X(F) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi Ft} dt$$

- ∞ to $-\infty$ integral is practically not relevant, as signals are finite and discrete (sampled)

Fourier Transform

- always an analog-to-digital conversion → **sampling**, i.e., finite number of samples
- → **Discrete Fourier Transform**



Discrete Fourier Transform

From Continuous to Discrete

Continuous Fourier Transform:

$$X(F) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi Ft} dt$$

Discrete Fourier Transform:

$$X_k = \sum_{n=0}^{N-1} x_n e^{\frac{-j2\pi kn}{N}}$$

- evaluated at the n-th sample for the k-th frequency

Attention:

Mathematical notation is not "correct" here, as the frequency is not continuous but discrete, so to represent the frequency in the discrete domain, we use

$$\frac{k}{N} \cong F \text{ and } n \cong t$$

Discrete Fourier Transform

$$X_k = \sum_{n=0}^{N-1} x_n e^{\frac{-j2\pi kn}{N}}$$

Expansion with substitution: $\frac{-j2\pi kn}{N} = b_n$

$$X_k = x_0 e^{-b_0 j} + x_1 e^{-b_1 j} + x_n e^{-b_n j}$$

Euler's formula: $e^{j\theta} = \cos(\theta) + j\sin(\theta)$

Using Euler's formula for Expansion:

- $X_k = x_0(\cos(b_0) + j\sin(b_0)) + x_1(\cos(b_1) + j\sin(b_1)) + \cdots + x_n(\cos(b_n) + j\sin(b_n))$
- $X_k = A_k + B_k j$

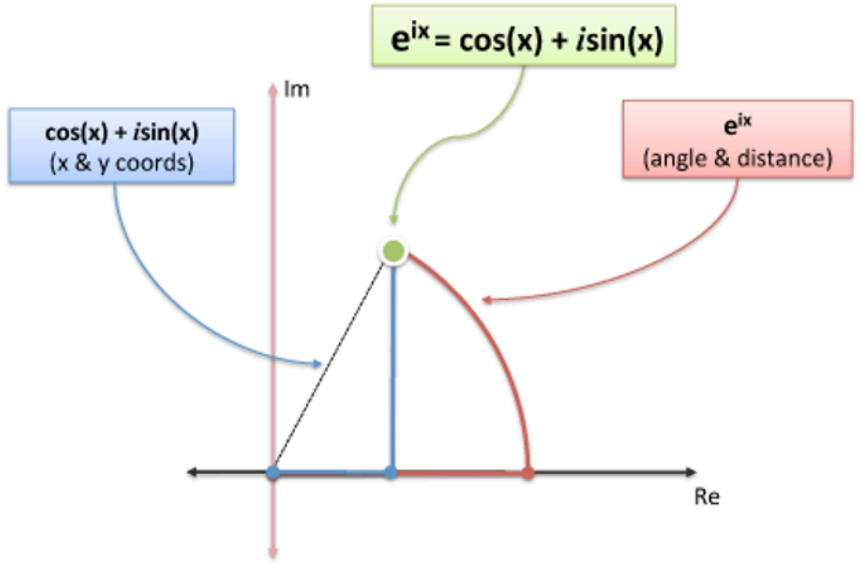
Discrete Fourier Transform

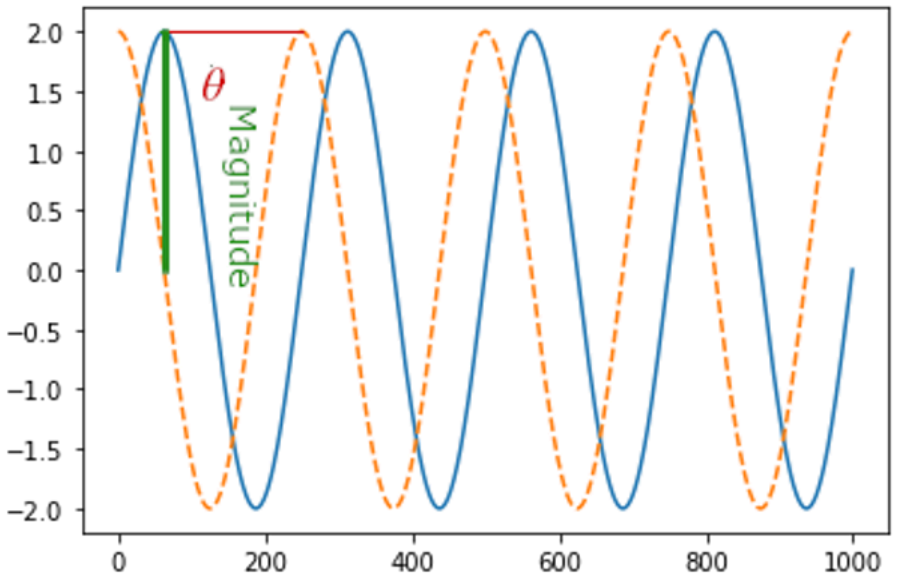
- $X_k = A_k + B_k j$
- Imaginary plane: A_k and B_k are the real and imaginary parts of the complex number X_k

Magnitude and Phase

- $|X_k| = \sqrt{A_k^2 + B_k^2}$
- **Phase / $\angle$:** $\theta = \arctan(\frac{B_k}{A_k})$

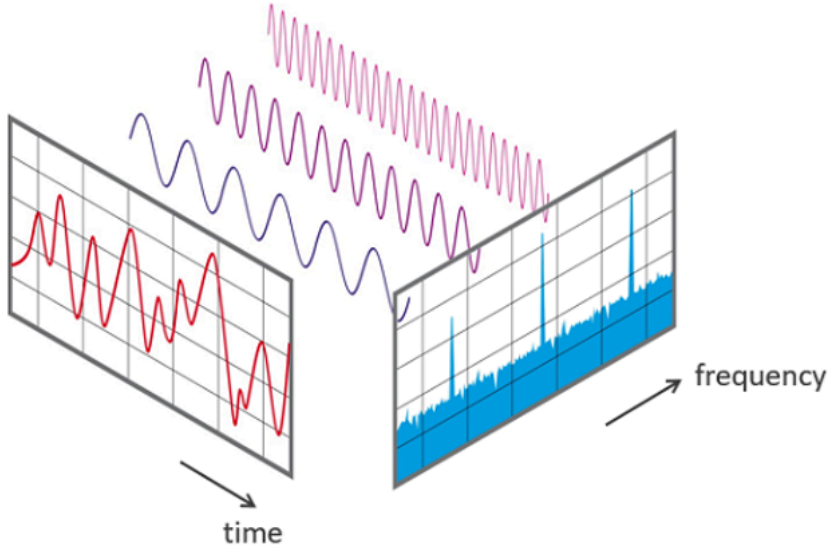
Go to: DFT – Example in jupyter notebook

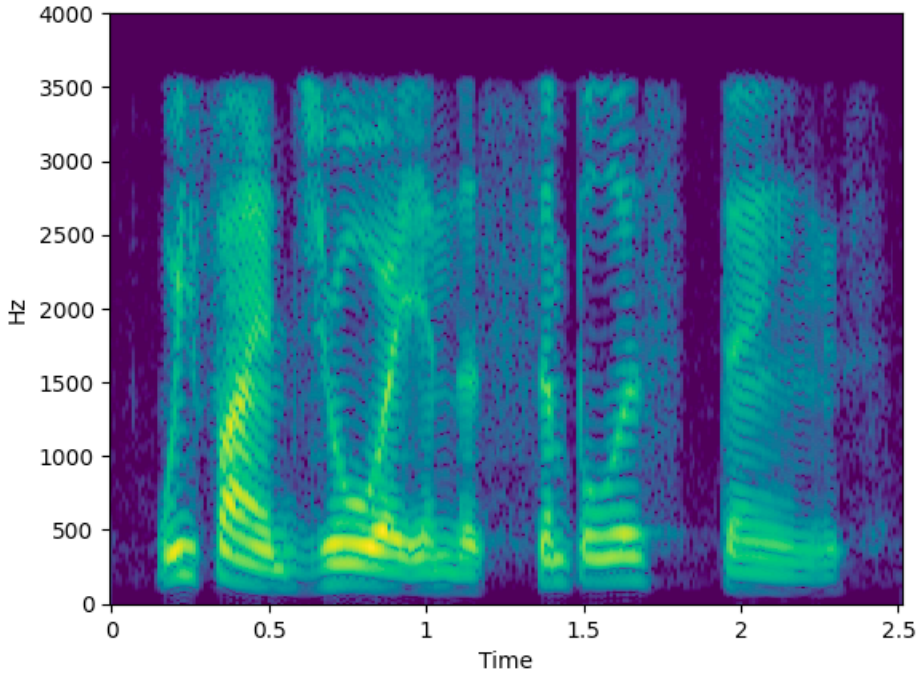




Spectrograms

- sound is disassembled as a series of sinusoids with varying amplitudes, frequencies, and phases
- the total amplitude (wav-from) is not very informative
- The spectrogram is a 3-D representation of the signal
  - amplitude (1-dim) of
  - each frequency (1-dim)
  - over time (1-dim)
- Created by applying the Fourier Transform to overlapping windows of the signal
- spectrogram defined by:
  - the window length
  - the overlap
  - windowing function
  - number of FFT points

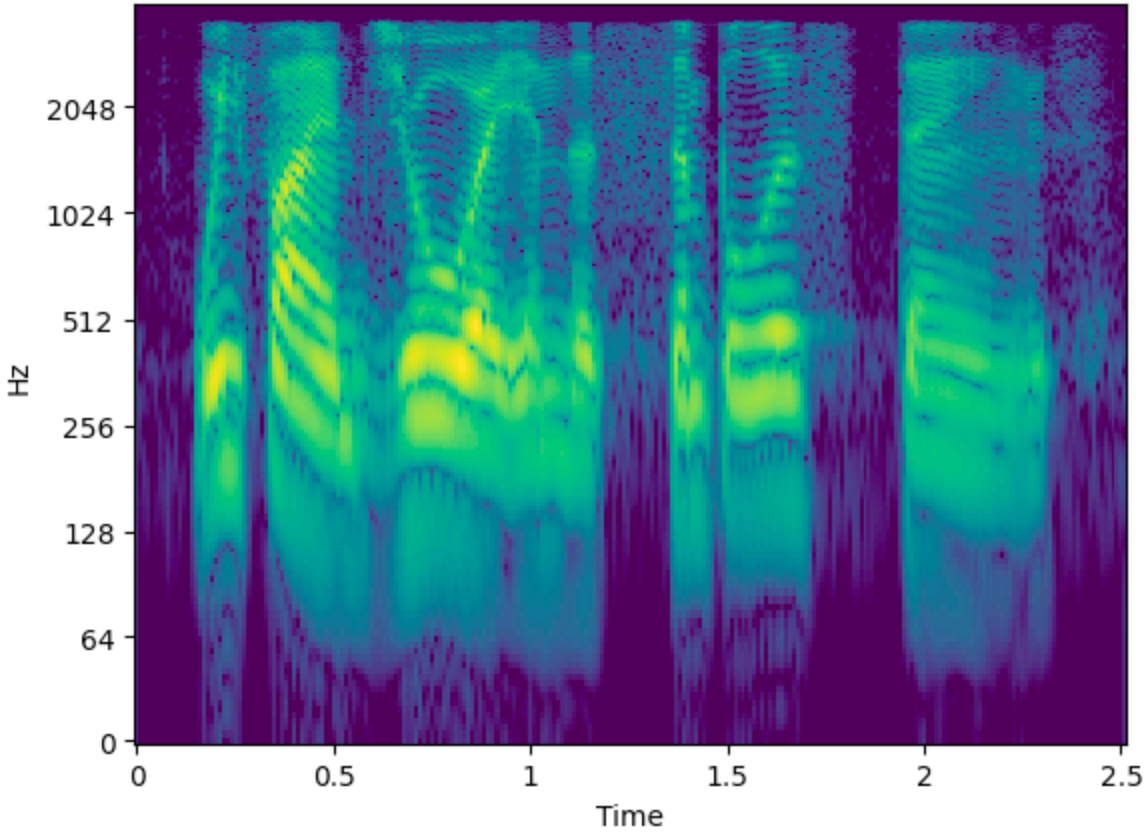




Spectrogram Variations

Log Spectrograms

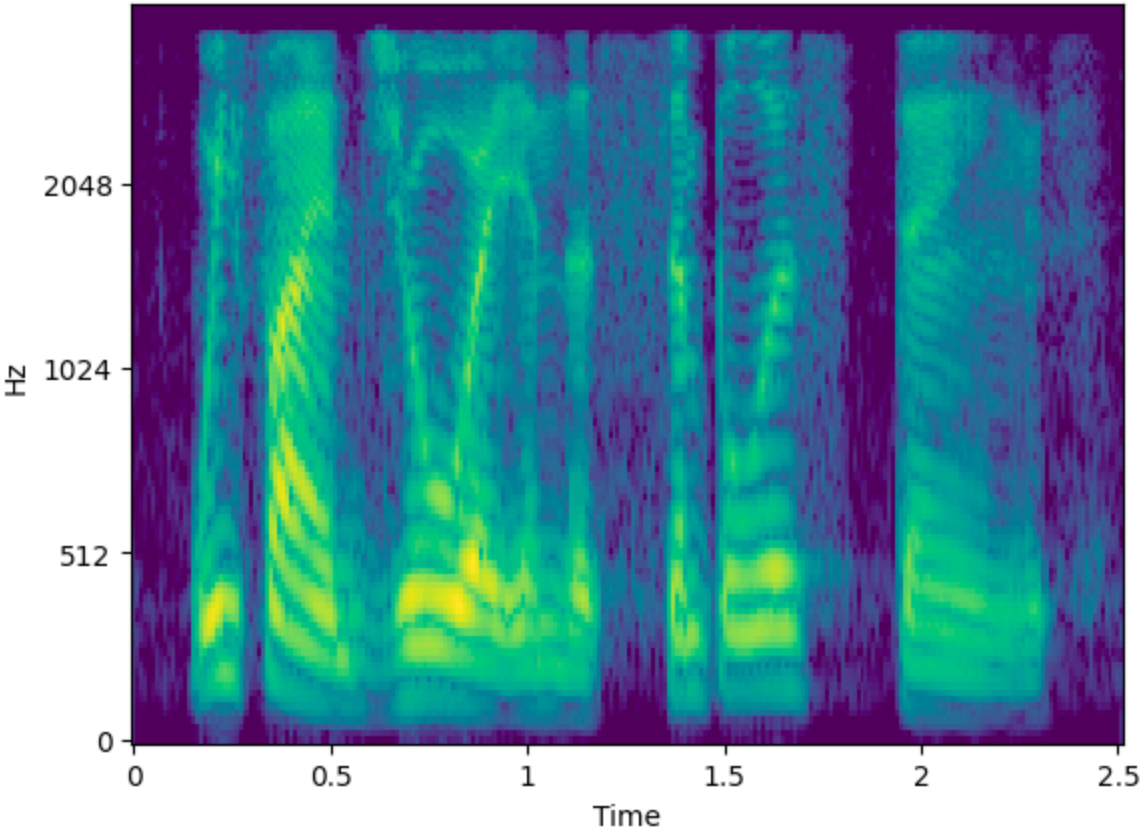
- A log spectrogram is a variation of the spectrogram
- where the amplitude values are converted to the log scale
  - allows better visualization of signals with a wide dynamic range
  - emphasizes the differences in lower amplitude values
  - which can be important for certain audio analysis tasks.



Spectrogram Variations

Mel Spectrograms

- A mel spectrogram is a variation where the frequency axis is transformed to the mel scale
- The mel scale approximates the human ear's response to different frequencies meaning it emphasizes important frequencies and cuts out less important ones (e.g. very high ones)
- makes spectrograms more suitable for tasks where human perception is a key factor
  - automatic speech recognition
  - audio recognition



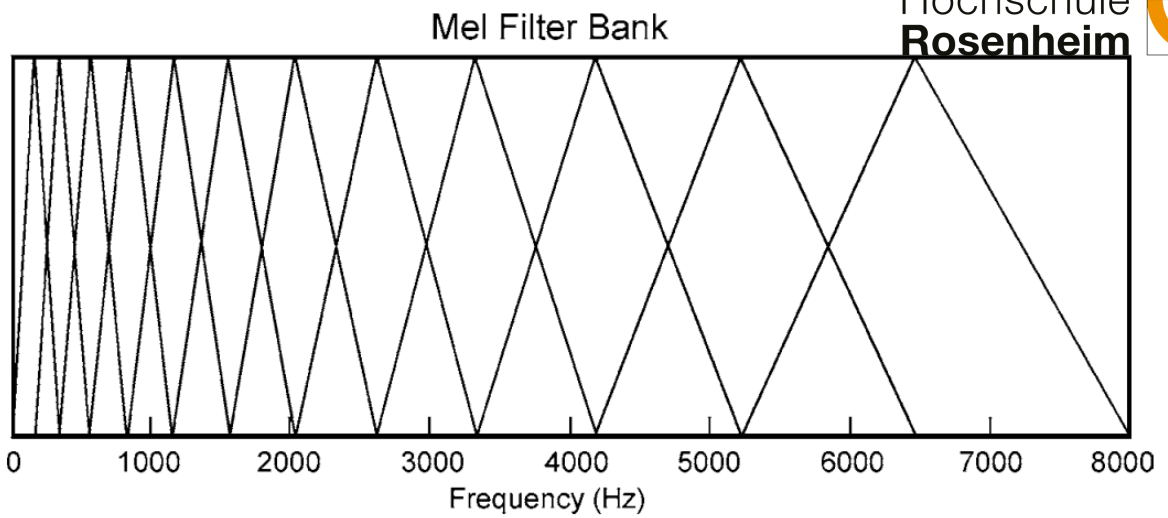
Spectrogram Variations

Mel Spectrograms

- Mel scale is a perceptually motivated scale that approximates the human ear's response to different frequencies.
- The scale is linear below 1 kHz and logarithmic above 1 kHz

Conversion from Hertz to Mel

$$M(f) = 2595 \log_{10} \left(1 + \frac{f}{700}\right)$$



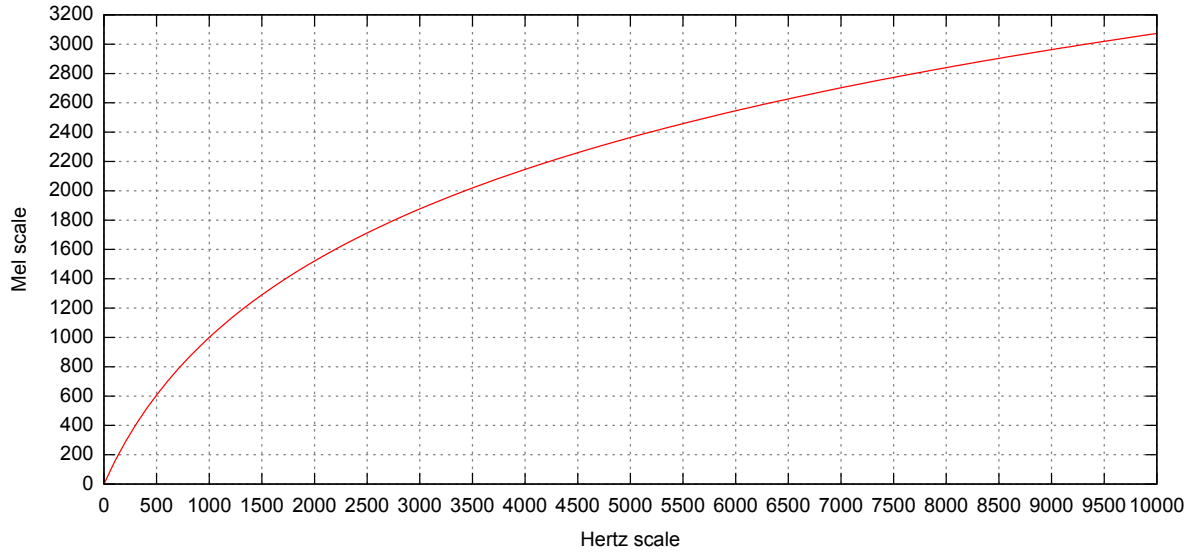
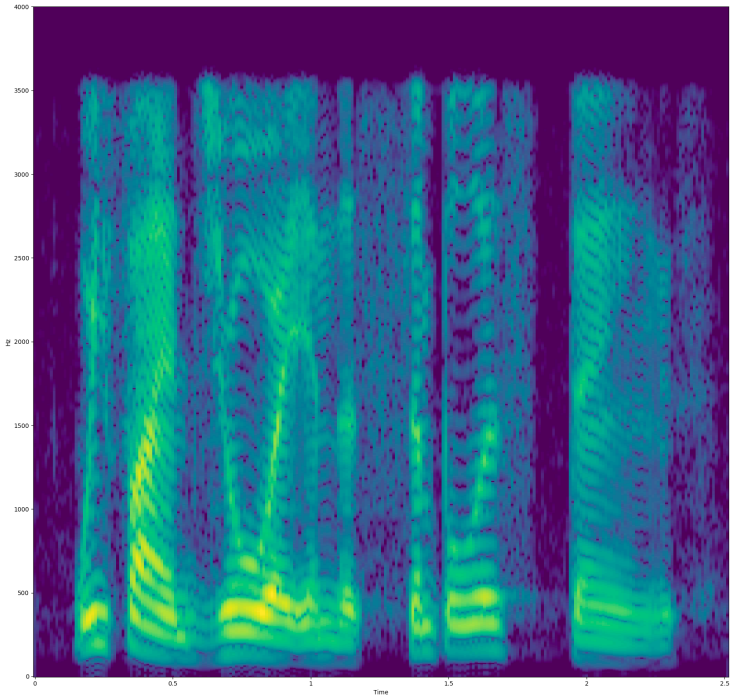
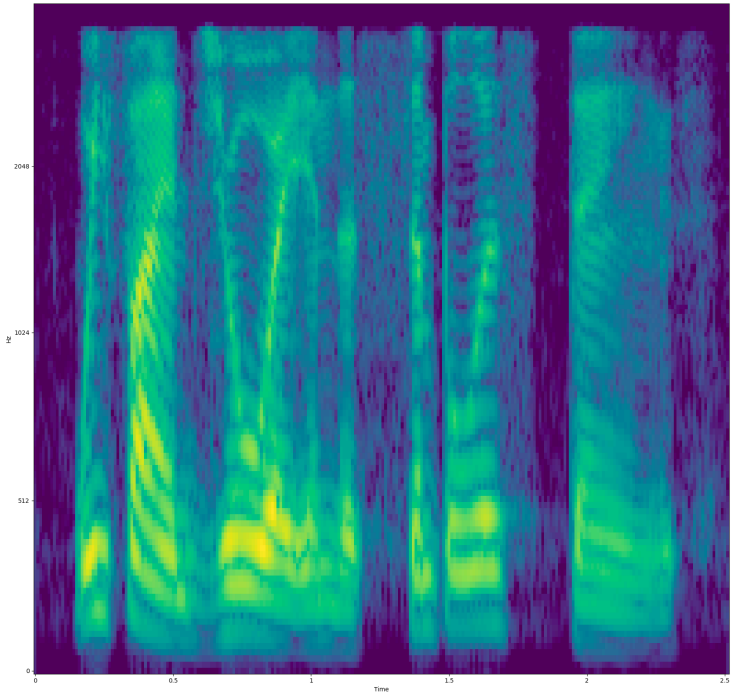


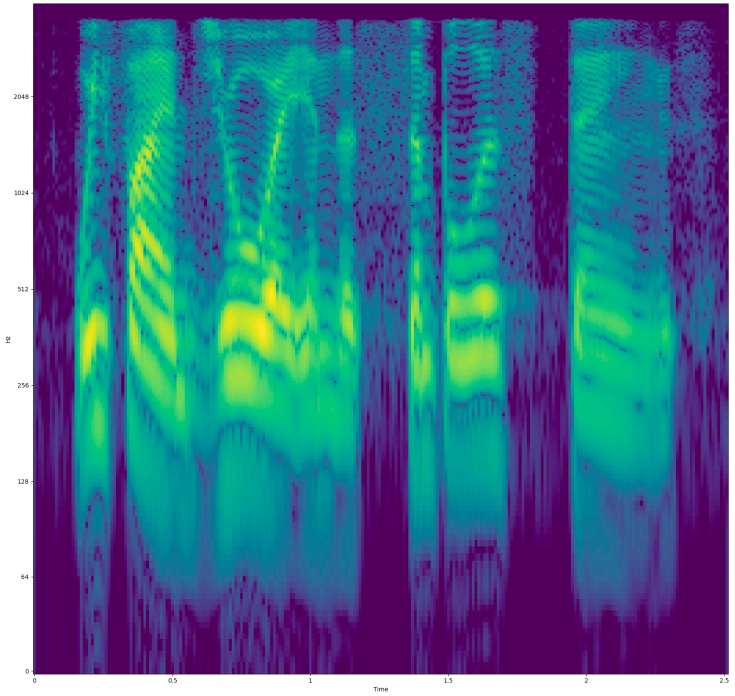
Image source: https://en.wikipedia.org/wiki/File:Mel-Hz_plot.svg

Spectrograms

From left to right: linear scale, mel-scale, log-scale







Mel-frequency cepstral coefficients (MFCCs)

- MFCCs are frequency-based features extracted from audio signals
- derived from the mel spectrogram
- modified by computing the discrete cosine transform (DCT) of the log-magnitude mel spectrogram
- MFCCs are designed to capture the timbral aspects of audio signals
- widely used in speech and audio processing.

Spectrogram Variations

Technische Hochschule Rosenheim

Deltas and Delta-Deltas

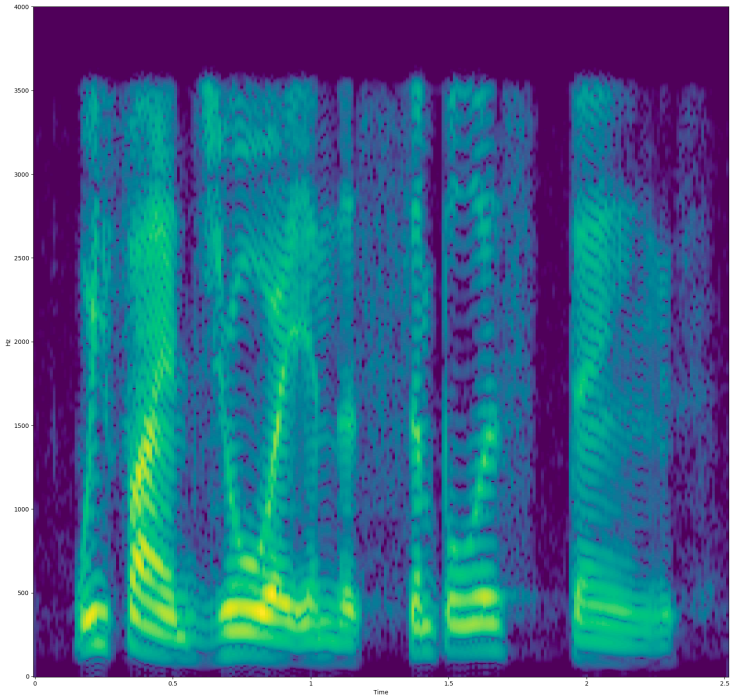
- Delta spectral features, also known as delta coefficients, represent the rate of change of the spectral features over time.
- Using the first and second derivatives (Δ, $\Delta\Delta$) of the spectrogram
- discrete approximation of the derivative per context window of consecutive frames
- typically used in conjunction with static spectral features to provide temporal information.
- Delta features capture the dynamic changes in the spectral features, providing additional context for
  - Speech recognition, phones have transitions.
  - Speaker recognition, to capture speaking style and intonation.
- The delta feature Δ_t at time t is calculated using the following formula:

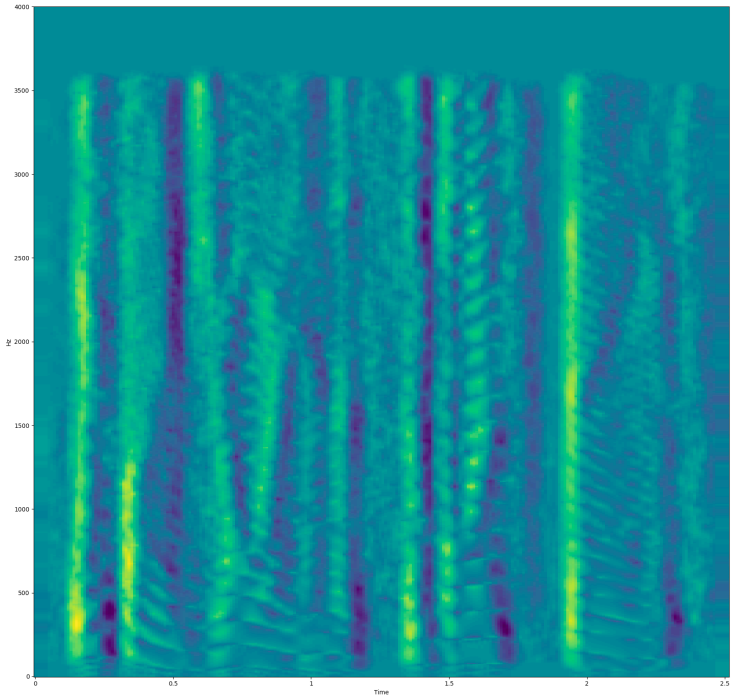
$$\Delta_t = \frac{\sum_{n=1}^{N} n \cdot (C_{t+n} - C_{t-n})}{2 \cdot \sum_{n=1}^{N} n^2}$$

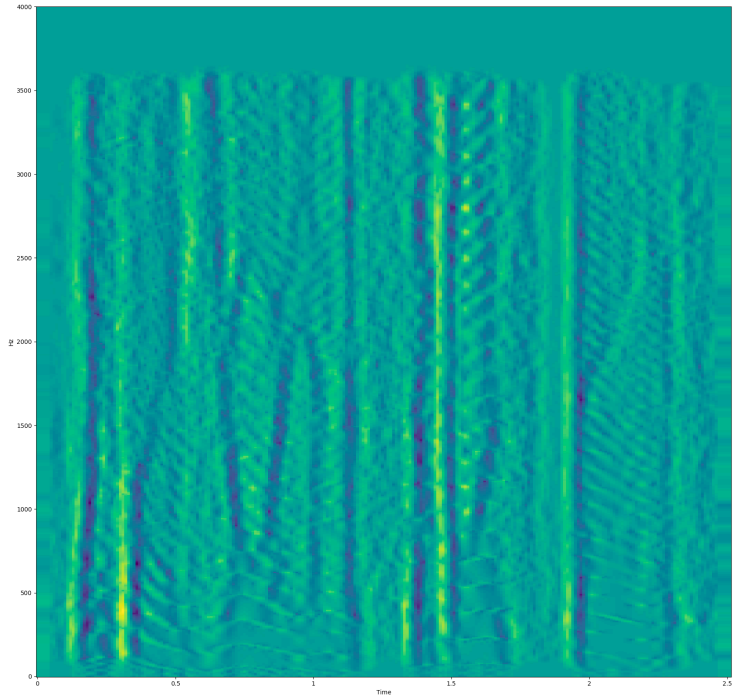
  - With:
    - Δ_t is the delta feature at time t
    - C_t is the static spectral feature at time t
    - N is the number of frames over which the delta is computed.
    - n is the index of the frame offset.
- The numerator computes the weighted difference between feature vectors at different time frames.
- The denominator normalizes the delta value

Spectrogram Variations

From left to right: Power Spectrogram, it's Deltas and Delta-Deltas







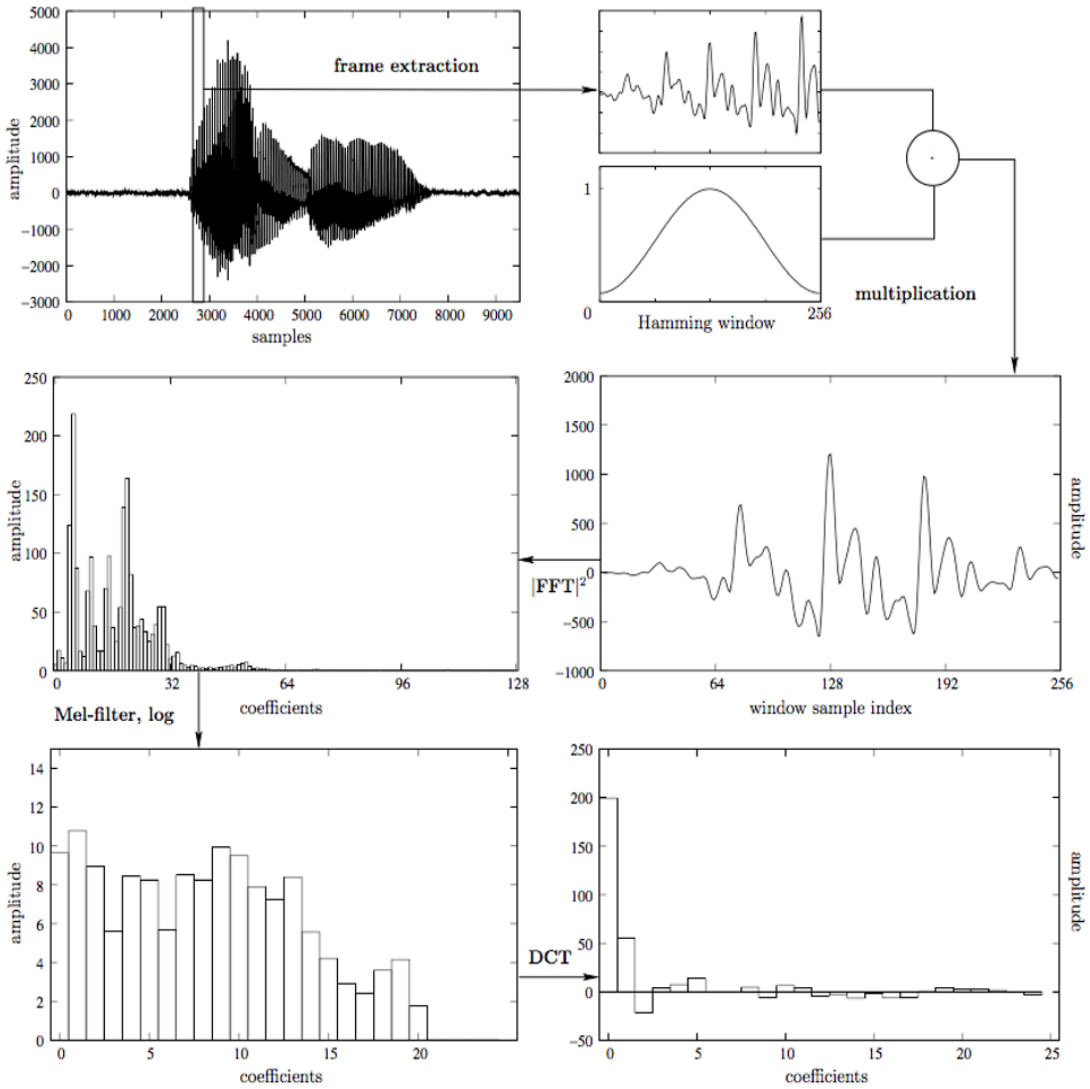
MFCC Features - Computation

Steps to compute MFCCs:

1. Compute the spectrogram of the audio signal.
2. Convert the spectrogram to a mel spectrogram.
3. Take the logarithm of the mel spectrogram.
    - This step transforms the log-mel spectral features into a more compact form
    - has a decorrelating effect
4. Apply the discrete cosine transform (DCT) to the log-mel spectrogram.

The resulting MFCCs are a more compact representation of the audio signal's spectral properties.

strong compression without much information loss



MFCC Features

Differences Between MFCCs and Spectrograms

- Spectrograms provide a detailed time-frequency representation of the audio signal
- MFCCs, are more compact, perceptually motivated representation.
- Spectrograms are useful for detailed analysis and visualization
- MFCCs are usually more suitable for machine learning tasks due to their reduced dimensionality and focus on perceptually important features

Applications of MFCCs

- MFCCs are widely used in various applications such as:
  - Speech recognition
  - Speaker identification
  - Music genre classification
  - Audio scene analysis
- Their effectiveness in capturing timbral characteristics makes them a preferred choice for many audio analysis tasks.

FFT-based features: Practical Spectrogram Examples in Python

- spectrogram
- log power spectrogram
- mel spectrogram
- amplitude to db

Libraries:

- scipy
- librosa
- torchaudio
- soundfile

Outlook: Data-based Features

Two paradigms: waveform-based and (spectral) feature-based

- i-vectors (GMM)
- ECAPA-TDNN
- x-vectors (TDNN and ResNet based)
- wav2vec 2.0

Outlook: Speech Recognition

Summary

- speech is a complex signal and much more than just spoken language
- speech is influenced by the speaker, the listener, the environment, and the task
- processing speech is a complex task due to the continuous nature of the signal and environmental influence
- spectral analysis is a tool to extract features from speech signals (FFT)
  - short-time analysis
  - overlapping context windows (frames)
- spectrograms are interpretable 3-D representation of the signal

"Speech is the most sophisticated behaviour of the most complex organism in the known universe!"

Moore, R. K. (2007). Spoken language processing: piecing together the puzzle. Speech Communication, 49, 418-435.

Sources

- heavily based on the Interspeech 2020 Tutorial Speech 101 by Prof. Roger K. Moore
- Denes, P.B., Pinson, E.N., 1993. The speech chain: the physics and biology of spoken language.
- Benesty, J., Sondhi, M.M., Huang, Y.A. (Eds.), 2008. Springer Handbook of Speech Processing, Springer Handbooks. Springer Berlin Heidelberg, Berlin, Heidelberg.